Prompt Optimization Pipeline Using Multi-Criteria Heuristic Evaluation and Transformer Fine-Tuning Algorithms

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(OPTIONAL)

The student may choose to include an acknowledgment page to thank those whom they wish to show gratitude to such as their parents, advisers etc. The content must be in English.

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Chapter 1

INTRODUCTION

Large Language Models (LLMs) are now widely used in academic and professional settings to generate content, analyze information, and support decision-making (Sahoo, 2024). However, their performance remains highly dependent on how prompts are structured. Small variations in wording, constraints, or context can significantly alter outputs, making prompt interaction unstable and difficult to standardize (Schulhoff, 2025). As a result, prompt refinement is increasingly treated as a structured optimization problem rather than an intuitive skill (Ramnath, 2025). Still, many optimization methods remain costly and sensitive to evaluation design in black-box API environments, limiting reproducibility (Pryzant, 2023).

In the Philippine context, the adoption of generative AI tools has increased rapidly. The country ranks among the top global users of ChatGPT (Mateo, 2025), and both students and educators actively integrate AI into academic tasks (Martinez, 2025). Despite widespread use, recurring issues such as inaccurate, redundant, and unreliable outputs persist (Cabuquin et al., 2024), and trust remains cautious due to hallucinations and bias (Amoozadeh et al., 2024). Institutional guidance and technical governance frameworks are still fragmented (Gabayan et al., 2025). These patterns suggest that while AI use is common, systematic mechanisms for ensuring consistent output quality remain underdeveloped.

Although prompt optimization and fine-tuning techniques have been developed, it remains unclear how these approaches can be integrated within a unified framework. Existing literature has not yet established standardized multi-criteria evaluation methods that comprehensively assess semantic stability and output reliability across tasks. Moreover, while automated optimization strategies improve semantic fidelity, there is limited understanding of how such methods can be designed to minimize computational costs, evaluation overhead, and sensitivity when applied to large language models. Additionally, how to achieve consistent, efficient, and stable AI performance through structured evaluation and model adaptation remains insufficiently explored.

To address these gaps, this study proposes a Prompt Optimization Pipeline Using Multi-Criteria Heuristic Evaluation and Transformer Fine-Tuning. The system applies structured heuristic scoring based on clarity, specificity, and semantic preservation to evaluate candidate prompts, then reinforces validated improvements through parameter-efficient transformer adaptation. By integrating evaluation and fine-tuning within a unified pipeline, the proposed framework aims to produce more stable, interpretable, and consistent outputs across tasks.

1.1. Background of the Study

The rapid integration of Large Language Models (LLMs) into academic and professional environments reflects a structural shift in how knowledge work and software development are performed. Recent demographic analyses indicate that AI-assisted coding and prompt-driven workflows have moved beyond experimental adoption into mainstream production use, particularly among users aged 25–34

(Cuofano, 2025). This widespread adoption spans students, educators, and professionals, reinforcing that AI prompting is now embedded in everyday workflows. However, widespread adoption does not guarantee output reliability, stability, or interpretability. The challenge is no longer whether AI systems are used, but whether their interactions are systematically optimized to deliver consistent, explainable results.

In Philippine education, AI usage has become normalized across year levels regardless of academic experience (Dela Rosa et al., 2024), yet prompt construction is often treated as an intuitive practice rather than a formally evaluated process (Hatmanto et al., 2024). While awareness and perceived usefulness are moderately high (Asirit & Hua, 2023), technical mechanisms for systematically improving interaction quality remain limited. Broader industry studies similarly reveal uneven AI readiness despite high levels of digital infrastructure (Quimba et al., 2024), reinforcing that adoption alone does not ensure effective or reliable utilization.

Globally, prompt engineering is increasingly treated as an optimization problem rather than a manual practice, with Automatic Prompt Optimization systems framing it as an iterative process of candidate generation, evaluation, and refinement (Ramnath et al., 2025). However, results depend heavily on how evaluation criteria are defined, while benchmarking remains fragmented and often prioritizes accuracy over robustness, clarity, interpretability, and efficiency (Zhang et al., 2025; Schulhoff et al., 2025; Sahoo et al., 2024). Although transformer fine-tuning methods such as LoRA enable parameter-efficient adaptation and reduced memory usage (Hu et al., 2021; Choi et al., 2025), they typically focus on parameter updates without integrating structured prompt-level evaluation. Moreover, both automated prompt optimization and

fine-tuning approaches introduce computational and hardware costs, revealing a gap between optimization design, evaluation consistency, and model adaptation.

1.2. Statement of the Problem

AI adoption in the Philippines has reached a relevant usage rate of 42.4%, and counting, doubling the global average in 2025 (Mateo, 2025). The widespread use of AI exhibits that more people are embracing the use of LLMs as a tool to excel in terms of performance and efficiency. Despite these benefits, there are still numerous issues being encountered, including redundant, inaccurate, and inadequate responses (Cabuquin et al. 2024). The cause of these problems is what other researchers call prompt brittleness (Syahputri et al. 2025), where small mistakes or variations in the user's prompt tend to cause inconsistent results and hallucinations. Given these challenges, the system aims to provide an automatic prompt optimization pipeline. By utilizing Multi-Criteria Heuristic Evaluation and Direct Preference Optimization (DPO) via QLoRA fine-tuning, the system aims to transform user prompts into precise and structured instructions without the overhead of massive, proprietary APIs.

Specifically, this study seeks to address the following questions:

1. What are the identified challenges of raw AI prompts, specifically prompt brittleness, lack of clarity and specificity. Additionally, what evaluations can serve as the respondents' verification of these challenges?
2. How can Multi-Criteria Heuristic Evaluation and parameter-efficient Transformer Fine-Tuning algorithms be integrated to develop a fully automated

pipeline capable of transforming raw inputs into structured, high-quality prompts?

3. How can the algorithmic performance and effectiveness of the DPO-aligned model in overcoming prompt brittleness be measured using defined heuristic metrics, expert reviews, and win-rate comparisons?
4. How can the developed prompt optimization system's overall software quality, and user acceptance be assessed utilizing the ISO 25010 framework and TAM?
5. How can the system's evaluation results be utilized to formulate actionable recommendations and institutional guidelines for the responsible AI use in educational settings?

1.3. Objectives of the Study

1.3.1. General Objective

The researchers aim to design, implement, and evaluate a fully automated prompt optimization system that rewrites raw inputs into higher-quality structured prompts by utilizing a Multi-Criteria Decision-Making (MCDM) heuristic engine to guide the supervision of preference datasets. This data is then used to fine-tune a Qwen-Instruct model via QLoRA and DPO, resulting in a parameter-efficient system deployable on memory-constrained hardware environments.

1.3.2. Specific Objectives

1. To identify the challenges of raw AI prompts, namely prompt brittleness, lack of clarity, and lack of specificity (Syahputri, 2025; Shelf, 2025), by using the

system's heuristic scorer and verifying them through the respondents' approval or disapproval ratings

2. To develop a fully automated prompt optimization pipeline capable of transforming raw inputs into structured prompts utilizing Multi-Criteria Heuristic Evaluation and parameter-efficient Transformer Fine-Tuning algorithms
3. To measure the algorithmic performance and effectiveness of the DPO-aligned model using predefined heuristic metrics, human expert reviews, and LLM-as-a-judge win-rate comparisons.
4. To assess the developed system's software quality using ISO 25010 framework and evaluate user acceptance using the Technology Acceptance Model (TAM) through structured surveys and statistical analysis.
5. To formulate institutional guidelines and actionable recommendations for responsible AI use in educational settings through the analyzed system evaluation results and prompt scoring data.

1.4. Theoretical Framework

1.4.1. Requirements Engineering and Garbage In, Garbage Out (GIGO)

Requirements engineering (RE) provides a direct lens for prompt optimization because both requirements and prompts are natural-language specifications intended to constrain and guide system behavior. Vogelsang (2024) argues that generative LLM practice shifts "from specifications to prompts," which elevates

precision, structure, and explicit constraints as quality properties. This matters because LLM use in RE and related tasks remains limited by evaluation, reproducibility, and validity challenges at scale (Norheim et al., 2024).

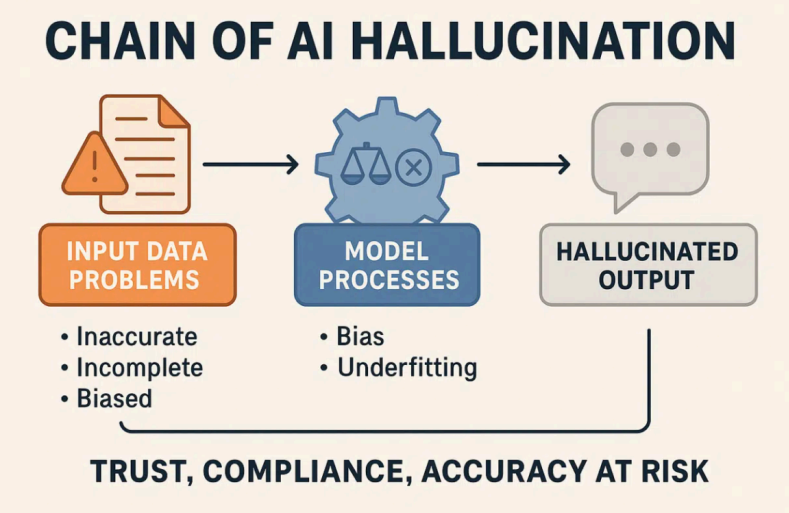

Figure 1. Chain of AI Hallucination as Part of the GIGO Principle

Note. Reprinted from Oksana Zdrok. (2024, May 2). "Garbage In, Garbage Out": the New Mantra for AI Implementation. Shelf. https://shelf.io/blog/garbage-in-garbage-out-ai-implementation/

In Requirements Engineering (RE), the "Garbage In, Garbage Out" principle emphasizes that poor-quality inputs lead to poor-quality outputs. Recent literature on Large Language Models (LLMs) supports this concept, demonstrating that poorly written text descriptions reduce a model's reasoning capabilities (Simmons & Vasa, 2023). Figure 1. illustrates this effect. The cycle begins with flawed input data, such as inaccurate, incomplete, or biased information, which creates a weak foundation. As the model processes this data, internal issues like bias and

underfitting further corrupt the pipeline, inevitably producing hallucinated outputs. This flawed cycle ultimately places trust, compliance, and accuracy at risk. Because an AI model's accuracy relies directly on its input quality, feeding it vague or ambiguous prompts significantly increases the chance of these errors (Shelf, 2025). To prevent these issues, prompt optimization should be treated as requirements-like refinement. By improving specificity, resolving ambiguity, and clearly defining constraints, outputs become more reliable and evaluable (Vogelsang, 2024; Liu et al., 2021; Shelf, 2025). This justifies two methods in the study, including first, treat prompts as engineered artifacts whose quality can be decomposed into assessable criteria, and second, enforce disciplined evaluation and decision procedures during iteration rather than relying on anecdotal outputs (Norheim et al., 2024; Liang et al., 2022).

1.4.2. Quantized Low-Rank Adaptation (QLoRA)

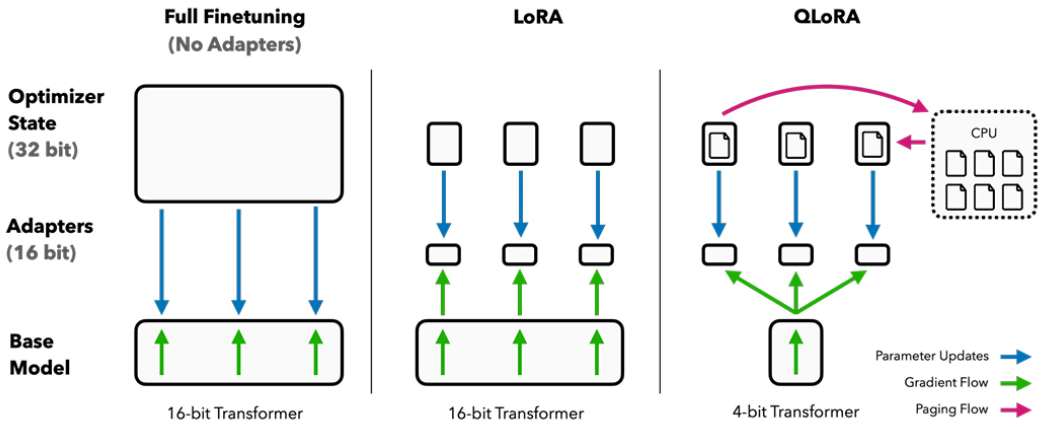

Figure 2. Different fine-tuning methods and their memory requirements.

Note. Reprinted from Dettmers, T., Pagnoni, A., & Zettlemoyer, L. (2023). QLORA: Efficient Finetuning of Quantized LLMs. https://arxiv.org/pdf/2305.14314

Adapting LLMs for specialized tasks could be highly resource-intensive. As shown in the left panel of Figure 2., Traditional full fine tuning approach involves updating every single parameter of a 16-bit transformer, resulting in a large 32-bit optimizer state that exceeds the limitations of usual consumer hardware. A way of mitigating these constraints is through Low-Rank Adaptation (LoRA) which is depicted on the center of the Figure 2, LoRA freezes the pre-trained model weights, denoted as $W_0 \in \mathbb{R}^{d \times k}$ and injects a subset of trainable rank decomposition matrices, called adapters into each layer. Mathematically, the forward pass is modified such that the updated weight matrix is $W' = W_0 + B\,A$ where $B \in \mathbb{R}^{d \times r}$ *and* $A \in \mathbb{R}^{r \times k}$, with rank $r \ll min(d, k)$ (Dettmers et al., 2023). During training, gradients only flow to the 16-bit adapters, therefore reducing the size of the optimizer state significantly. As demonstrated on the right panel of the figure, QLoRA further advances this technique by quantizing the frozen base weights to a 4-bit data type, which then drastically reduces the model's memory utilization as compared to the 16-bit baseline in LoRA. Additionally, the use of a paging flow, depicted by pink arrows, allows the 32-bit optimizer state to be offloaded to the CPU, preventing GPU from running out of memory. This combination of 4-bit quantization and memory paging allows models with large parameter values to be fine-tuned for tailored and specialized tasks on consumer hardware without drastically sacrificing performance (Dettmers et al., 2023; Hu et al., 2021).

1.4.3. Heuristics in Multi-Criteria Decision-Making

The evaluation of generated text within an autonomous optimization pipeline requires a structured approach to rank competing outputs. Traditionally, alignment methods utilize a computationally heavy, separate neural reward model to exhaustively analyze and score these outputs. To bypass this immense information processing requirement, this system grounds its autonomous scoring engine in the established theory of Multi-Criteria Decision-Making (MCDM), specifically leveraging fast and frugal heuristics. Recent research in decision processes demonstrates that applying heuristics in multi-criteria environments allows systems to drastically reduce computational effort while maintaining the accuracy of the final alternative rankings (Methling et al., 2022). Heuristics achieve this efficiency by selectively focusing on a limited set of the most critical attributes of an alternative, rather than processing all available analytical data. In the context of automated prompt engineering, the system evaluates various generated prompt transformations (the alternatives) using a deterministic heuristic algorithm, ensuring highly efficient generation of preference data.

MCDM theory dictates that for a fast and frugal heuristic to be effective, it must operate on a rigorously defined set of attributes that accurately reflect the decision objective (Methling et al., 2022). Within this prompt engineering framework, Clarity, Specificity, and Semantic Preservation are amongst the fundamental attributes that are empirically consistent. Clarity and Specificity act as the primary heuristic cues for structural prompt quality, measured through syntactic dependency

parsing and constraint identification (White et al., 2023). Semantic Preservation acts as the boundary constraint, utilizing cosine similarity to ensure that the heuristic does not highly rank a prompt that has deviated from the raw input's original intent. By restricting the mathematical evaluation strictly to these criteria, the system operationalizes the fast and frugal paradigm. It rapidly calculates a multidimensional score for each prompt variation, autonomously scoring them to construct the winning y_W and losing y_L preference pairs required for the Direct Preference Optimization loss function.

1.4.4. Direct Preference Optimization (DPO)

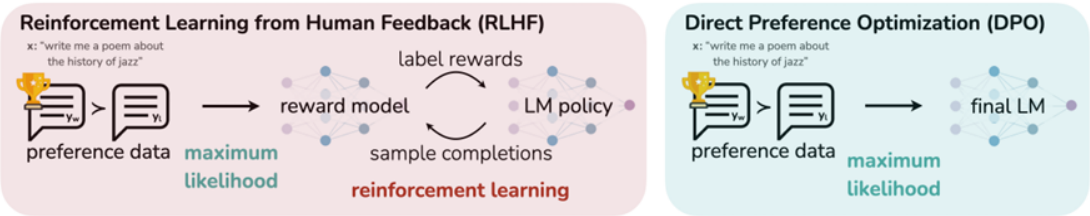

Figure 3. DPO Optimizes for Human Preferences While Avoiding Reinforcement Learning

Note. Reprinted from Rafailov, R., Sharma, A., Mitchell, E., Ermon, S., Manning, C., Finn, C., & Cz Biohub. (2024). *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*. https://arxiv.org/pdf/2305.18290

Aligning language models to exhibit specific generative behaviors has historically relied on Reinforcement Learning from Human Feedback (RLHF). As illustrated in the left panel of Figure 3, RLHF utilizes a much more complex, multi-stage pipeline that requires first training a separate reward model via maximum likelihood,

followed by iteratively optimizing the language model policy through a reinforcement learning loop that samples completions and assigns reward labels. Direct Preference Optimization (DPO), on the other hand, as depicted in the right panel of Figure 3, shifts this paradigm. DPO eliminates the need for a separate reward model and a subsequent reinforcement learning phase. Instead, it formulates the alignment process as a single classification problem, utilizing maximum likelihood to directly optimize the final language model against a dataset of preference pairs, containing a preferred or winning response y_w and a rejected or losing response y_l. By mathematically demonstrating that the language model can act as its own implicit reward model, DPO achieves stable alignment with significantly reduced architectural overhead (Rafailov et al., 2023).

This theoretical framework serves as a foundation for the proposed automatic prompt engineering pipeline. The objective of this study is to automatically transform raw prompt inputs into optimized, heuristically-guided prompts using a Qwen-Instruct base model. Implementing a traditional RLHF pipeline for this task would introduce a significant computational overhead and require continuous, dynamic reward evaluations that are difficult to fully automate. By grounding the alignment methodology in DPO theory, the system can efficiently leverage deterministically generated preference data. Specifically, the system's Multi-Criteria Heuristics engine will autonomously evaluate generated prompt variations, generating the necessary preference data by explicitly defining the winning and losing prompts based on the defined attributes. DPO provides the established theoretical justification to map these heuristically scored pairs directly to the

optimization of the Qwen-Instruct model. This approach ensures the model learns the intricate syntax of optimal prompt generation directly from the heuristic feedback, entirely bypassing the instability and resource constraints associated with reinforcement learning algorithms (Rafailov et al., 2023).

1.5. Conceptual Framework

The framework functions as a guide to understand and clearly define the flow of the system. Using the input-process-output (IPO) model, the framework provides a brief view of the system from raw inputs to the optimized prompt. The framework is designed to address the gap in this field.

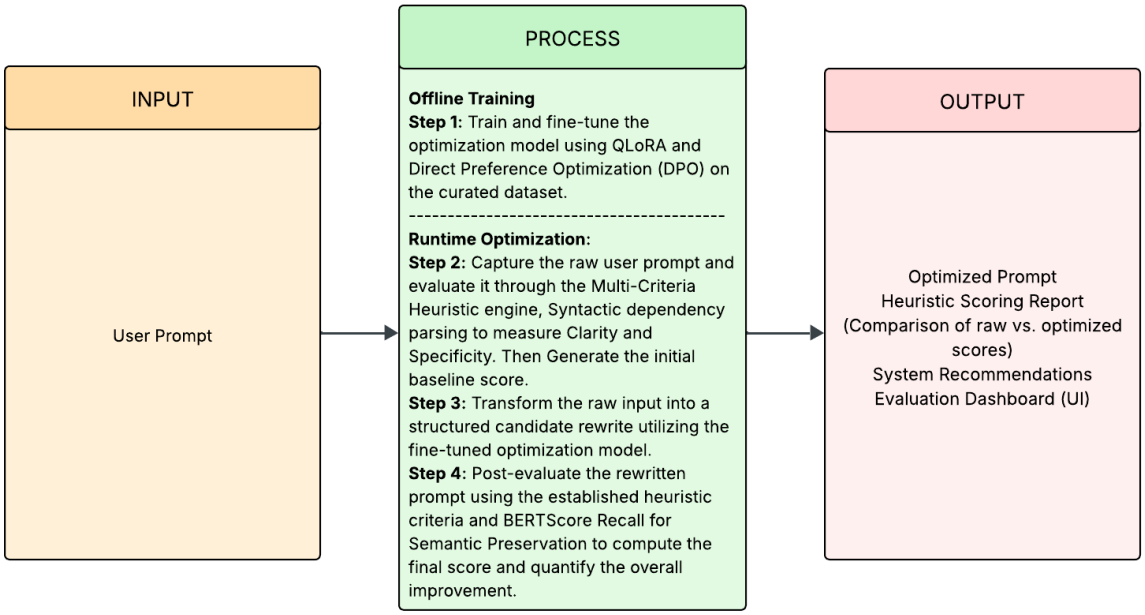

Figure 4. IPO Block Diagram

1.5.1. Input

The system's sole input is the User Prompt. This represents the initial, unoptimized natural language instruction or query provided by the end-user during live runtime operations.

1.5.2. Process

The Process block is systematically divided into an offline training phase and a continuous, step-by-step live runtime optimization sequence.

Step 1: The optimization process begins offline, where the model is trained and fine-tuned using Quantized Low-Rank Adaptation (QLoRA) and aligned with human preferences via Direct Preference Optimization (DPO) on a curated dataset. This establishes the system's foundational rewriting capabilities.

Step 2: Once deployed in the live runtime environment, the system captures the raw user prompt and evaluates it through the Multi-Criteria Heuristic engine. During this step, the system employs syntactic dependency parsing to mathematically measure the prompt's Clarity and Specificity, ultimately generating an initial baseline quality score.

Step 3: The system then passes the raw input to the fine-tuned optimization model (developed in Step 1), which autonomously transforms the input into a highly structured candidate rewrite.

Step 4: Finally, the system conducts a post-evaluation of the rewritten prompt using the established heuristic criteria. In this stage, it specifically integrates BERTScore Recall to strictly enforce Semantic Preservation, ensuring the

model did not hallucinate or lose the user's original intent, to compute the final score and quantify the overall improvement.

1.5.3. Output

The primary output is the Optimized Prompt, a highly structured instruction ready to be deployed to external language models. Accompanying this is a detailed Heuristic Scoring Report, which provides a mathematical comparison between the raw and optimized scores. Based on these evaluations, the system also generates specific System Recommendations to guide the user on prompt formulation. All of these outputs are aggregated and visually presented to the user through an interactive Evaluation Dashboard (UI).

1.6. Significance of the Study

The proposed system goes beyond automated prompt correction, serving as an educational tool that guides users in understanding the essential criteria of producing high-quality prompts. By comparing the results and scores of the raw and optimized prompts, the user can utilize the system as an educational tool to observe and discover the patterns and elements that constitute a good prompt. This process highlights the need for proper input formulation, which aligns with the concept of the Garbage In, Garbage Out (GIGO) principle, which claims that the quality of output produced by the AI system is directly proportional to the quality of the prompt (Shelf, 2025). This process also helps to ensure that AI is not misused by reducing the ambiguous prompts that often lead to hallucinations. This way, the system encourages the responsible use of

AI without relying on it blindly and instead uses it effectively in academic and professional settings.

Students. As the primary end-users and the targeted respondents for this study, more than half of the Filipino students tend to rely on Artificial Intelligence (Martinez, 2025). This study addresses the trial-and-error involved in prompting to obtain high-quality, relevant answers from the LLM for a specific task. Furthermore, students would be able to compare the structural difference between the weak and strong prompts.

Technical Writers. This study provides regulation in documentation and communication workflows. Technical writers often struggle with the inconsistencies of AI outputs; this study acts as an automated auto-correction for prompts. By ensuring consistency and quality of the optimized prompts is transformed from the raw input that the writer made. If the patterns are observed, they could also serve as a guide to avoid repeated, tedious testing and allow writers to focus on their task rather than on prompt manipulation.

General End-Users. This study would allow end-users to be more appreciative and maximize the potential of AI. By acting as a translator for their prompts, end-users can circumvent prompt brittleness, which is noise or ambiguous instructions in a prompt. It would allow the user to avoid hallucinations and access high-performance AI outputs by giving them prompts that are unrestricted by one's proficiency in prompt optimization automatically

Future Researchers. This research serves as an initial schematic for future work in automatic prompt optimization systems. Future scholars can build, tweak, experiment,

or adapt the specific heuristic weights leveraged in the study, as well as the fine-tuning methodology for a specific domain utilization. It produces a scalable developmental design that motivates innovation for specialized linguistic tasks.

1.7. Scope and Delimitations

Scope

The focus of this research is the development and assessment of the proposed prompt optimization pipeline to counter the problem of prompt brittleness using heuristic evaluation and DPO. The scope of this research will be determined by the following parameters: The system combines a Multi-Criteria Heuristic Evaluation component with a transformer-based rewriting engine suited to optimize instructional task-type prompts, specifically tailoring for Code generation and academic assistance prompts for testing. The optimization process utilizes Quantized Low-Rank Adaptation (QLoRA) for efficient fine-tuning and a Direct Preference Optimization (DPO) training pipeline. The optimization will be measured based on a specified weighting strategy, which highlights the role of clarity, specificity, and semantic preservation. Deployment would be a web-based architecture utilizing FastAPI for the backend and React for the frontend. The system would be evaluated quantitatively via the ISO 25010 and the Technology Acceptance Model (TAM)

Delimitations

However, the system would be defined within the context of the single-turn prompt optimization for instructional task-type prompts. Repeated refinements are currently not

supported. Moreover, the heuristic scoring and DPO training would also be confined to the given domain-specific datasets. The system's performance in other domains is not guaranteed. Additionally, the system will be deployed exclusively as a web-based application and will not be developed as a mobile application for Android or iOS platforms. The expected output from the system is within the boundaries of the English language only. Concepts beyond the model's pre-trained knowledge are outside the scope of the study. The research also recognizes that quality in a prompt can be subjective in nature. Although the system mathematically optimizes prompts based on objective heuristic criteria, it is possible that the output does not always meet the user standards, hence requiring human validation to ensure that it is in line with user preferences.

1.8. Definition of Terms

Backpropagation – The training process used in neural networks, where the model adjusts its parameters by learning from its errors during fine-tuning.

Black-Box Testing – A testing method that checks whether the system produces correct outputs based on given inputs without examining the internal code.

Causal Language Model (Causal LM) - A type of language model that generates text by predicting the next word based on the words that come before it.

Clarity – The extent to which a prompt clearly states the task, role, and expected output without confusion or ambiguity.

Cosine Similarity – A mathematical method used to measure how similar two pieces of text are in meaning.

Direct Preference Optimization (DPO) – A fine-tuning method that improves a language model by training it to prefer better prompt versions over weaker ones.

Fine-Tuning - The process of further training a pre-trained language model so it performs better on a specific task.

Heuristic Evaluation Module - The system component that calculates a quality score for a prompt using predefined evaluation criteria.

Heuristic Scoring Framework - A deterministic scoring mechanism that quantitatively evaluates prompt quality using predefined attributes.

Kullback-Leibler divergence (KL-Divergence) - A non-symmetric metric that measures how different one set of probabilities is compared to the actual or correct set of probabilities.

Multi-Criteria Decision Making (MCDM) - A decision-making approach that evaluates several alternatives using multiple criteria to determine the most suitable option.

Large Language Model (LLM) – A deep learning model trained on large-scale textual data capable of generating and understanding human language.

Low-Rank Adaptation (LoRA) - A parameter-efficient fine-tuning technique that freezes the base model weights and injects small, trainable rank-decomposition matrices into the transform layers.

Post-Scoring Validation – The process of re-evaluating an optimized prompt using the same heuristic criteria to measure improvement.

Preference Pair Dataset – A structured dataset containing preferred and rejected prompt variations used for DPO training.

Prompt Brittleness – The instability of model outputs caused by minor variations in prompt wording or structure.

Prompt Optimization - The system component that rewrites and improves a user's original prompt.

Prompt Optimization Pipeline – The complete system that evaluates, rewrites, and fine-tunes prompts to improve their quality.

Quantized Low-Rank Adaptation (QLoRA) - An improved version of LoRA that reduces memory usage by compressing model weights during fine-tuning.

Qwen-Instruct Model – The base transformer language model used in this study for prompt rewriting and fine-tuning.

React – The frontend framework used to display results and allow user interaction in the web interface.

Semantic Preservation – The degree to which the optimized prompt keeps the original meaning and intent of the user's input.

Specificity – The extent to which a prompt includes clear details and constraints to guide accurate responses

Chapter 2

REVIEW OF RELATED LITERATURES

This chapter presents a review of the relevant literature related to the present research. It includes both foreign and local literature that provide theoretical and empirical support to the variables investigated. The synthesis highlights gaps in previous research, which justify the conduct of the present study.

2.1. Local Related Literatures

2.1.1. The Generative Shift: A Systematic Review on Generative AI Use and Policies in the Philippines

Gabayan et al. (2025) conducted a systematic review of literature and institutional documents addressing the adoption, use, and governance of generative AI in Philippine higher education. The study synthesized academic publications, policy papers, and official university guidelines published between 2021 and 2025, using thematic analysis to identify patterns related to pedagogical value, ethical concerns, and institutional responses. The review identified a nationwide trend toward increased use of generative AI tools, such as ChatGPT, in academic writing, research assistance, and self-directed learning, alongside persistent concerns about academic integrity, authorship, and unequal access. While some universities have begun issuing

local AI guidelines, the absence of a unified national policy has led to fragmented and inconsistent practices across institutions. The review underscores that generative AI adoption is widespread but largely unmanaged from technical and evaluative perspectives, with little attention to structured mechanisms that ensure responsible, high-quality AI outputs. The authors call for coherent, context-sensitive governance frameworks that balance innovation with accountability and emphasize the need for systematic approaches to guide the ethical and effective use of generative AI in higher education.

2.1.2. Pinoys top ChatGPT users worldwide.

According to Mateo (2025), the Philippines ranks sixth globally in ChatGPT usage, indicating a high level of AI adoption. It is stated in the Digital 2026 report that the global average usage of the said LLM was 26.5%, but in the Philippine setting, it was nearly doubled, with 42.4% usage in the country. The obvious high usage rate highlighted in the article is a significant sign of reliance across different facets of sectors in Philippine society, ranging from health, fitness, and self-care to mathematical calculations.

2.1.3. More Filipino students now using AI for learning

Martinez (2025) reported from the Business World website that the quick adoption of AI in education in the Philippines is evident. More than half of Filipino students use artificial intelligence (AI) for academic assignments, according to the company that created the Canvas learning management system. BusinessWorld has the chance to exclusively interview Ryan Lufkin, the vice-president of Global Academic Strategy at

Instructure, and he told them that Filipino students are already actively using AI tools in their learning. Citing their own report in 2025, Lufkin said, “63% of Filipino students use generative AI chatbots like ChatGPT to generate texts, while 58% use it for translation tasks”. Lufkin also added that 55% of student asks AI-platform for guidance in understanding complex concepts. while 52% rely on them to summarize academic articles. According to Lufkin, it’s not just students who are utilizing LLMs but teachers as well, so Lufkin recommended that educators should also be literate in using AI. The same study revealed that “71% of educators in the country create learning materials and 65% generate quiz questions and assignments with the help of AI”. Additionally, “56% of teachers use AI to create personalized or adaptive learning materials for students, 37% use it for seminar or tutorial plans, and 31% use it for marking transcripts.”

2.1.4. Adoption of Artificial Intelligence Technologies in the Philippine Construction Industry: A Review of Literature

Santos and Jocson (2024) conducted a systematic review of literature examining the adoption of artificial intelligence in the Philippine construction industry, synthesizing studies published between 2014 and 2024 using a PRISMA-guided approach. Their review identifies productivity improvements, cost efficiency, safety enhancements, and sustainability as the primary drivers of AI adoption, particularly through decision support systems, predictive analytics, and automation tools. Despite these benefits, the authors emphasize that AI adoption remains uneven in the Philippine context due to persistent barriers, including limited data quality, high implementation costs, a shortage of skilled professionals, ethical concerns, and inadequate digital

infrastructure. The study does not present empirical validation; instead, it highlights systemic and organizational gaps that hinder effective use of AI. The authors recommend strengthening digital readiness, investing in workforce training, and establishing strategic and ethical governance frameworks to support broader and more responsible integration of AI across industries.

2.2. Foreign Related Literatures

2.2.1. Prompt Engineering in Clinical Practice: Tutorial for Clinicians

In a recent study tackling the ways to prompt properly for a better outcome, Liu et al. (2025) made 7 prompting techniques that are under 5 core principles: explicitness and specificity, contextual relevance, iterative refinement, ethical considerations, and evidence-based practices. To maximize the potential of LLMs and to avoid hallucinations and/or biases, the researchers provided 7 prompting techniques, mostly catering to clinical use. The first one is Zero-shot prompting, which relies on the LLM's pretrained knowledge but offers flexibility in general queries. Second, One-shot prompting requires a single exemplar to get the desired output. Third, few-shot prompting, which uses 2-5 structure examples to guide an LLM's output with a specific clinical task. Fourth, Chain-of-Thought must include step-by-step reasoning to establish the proper logic to achieve better output. Fifth, Self-consistency prompting improves the response by generating different outputs and selecting the most consistent one through comparison. Sixth, Generated Knowledge Prompting is a two-stage process where it would lay out the relevant and factual knowledge first, and

then apply that knowledge to address the clinical query. Lastly, Meta-prompting consists of two prompts, the “meta-prompt” and the “base-prompt”, where the meta-prompt refines and evaluates the base-prompt for better clarity, scope, and relevance. Regarding the future direction, they suggested a more adaptive and context-aware prompting, multimodal prompting, and personalization and accessibility.

2.2.2. "Garbage in, garbage out": How to stop your AI from hallucinating.

A recent publication by Shelf (2025) explores the operational causes and strategic mitigations of artificial intelligence hallucinations. Garbage in, Garbage out is a principle that dictates how an AI model’s accuracy is directly influenced by its input quality. In the context of prompt engineering, feeding a model with vague or ambiguous prompts heightens the risk of hallucinations. Just as AI systems generate faulty predictions when trained on inaccurate, incomplete, or irrelevant data, these systems also produce low quality outputs when guided by low quality instructions. The author further suggests crafting structured prompts that contain clear instructions, specific constraints, and proper context. Thus, applying clear guidelines to inputs prevent models from making meaningless associations and ensure generation of reliable and accurate responses.

2.2.3. A Systematic Survey of Automatic Prompt Optimization Techniques

Ramnath et al. (2025) presented a systematic survey of automatic prompt optimization techniques, framing prompt optimization as a formal search problem that seeks to identify prompts maximizing task performance under a defined

evaluation metric. The survey synthesized a wide range of studies and proposed a unifying framework describing the core components of prompt optimization systems, including seed prompt selection, feedback generation, candidate prompt creation, candidate selection, and iterative refinement. The review identified a consistent trend toward black-box optimization methods due to their model-agnostic nature and interpretability, while also highlighting challenges such as evaluation cost, sensitivity to semantically similar prompts, and the absence of standardized benchmarks. The authors emphasize the need for structured optimization loops driven by explicit fitness criteria and controlled search strategies, and recommend that future research prioritize efficiency, robustness, and generalizable evaluation practices across tasks.

2.2.4. The Prompt Report: A Systematic Survey of Prompt Engineering Techniques

According to Schulhoff et al. (2025). There is a conflicting terminology and weak ontological understanding about what constitutes a prompt and an effective prompting technique, which is why they conducted a systematic review grounded in PRISMA via a curated keyword set and combining human annotation with GPT-4-assisted screening to produce a dataset of 1565 relevant papers, prioritizing hard, task-agnostic prefix prompts. Their work developed a standardized vocabulary and a taxonomy of 58 text-only prompting techniques, extended to multilingual and multimodal variants, and supplemented the review with benchmarking and applied case studies. Their findings emphasized that evaluation remains variable and unstandardized and that technique performance can be highly sensitive to meaning-preserving changes in prompts, implying limited transferability across

models and datasets. Additionally, a case study demonstrated that automated prompt optimization via DSPy could outperform fully manual prompt engineering on a real-world classification objective. Moreover, the authors recommended problem-driven evaluation practices, beginning with simpler techniques, and situating new methods within the proposed taxonomy using ecologically valid case studies to improve comparability and long-term methodological accumulation.

2.2.5. Unlocking the Potential of the Prompt Engineering Paradigm in Software Engineering: A Systematic Literature Review

A similar study was made by Syahputri et al. (2025), but for the Software Engineering field. The study defines “prompt brittleness” as the main problem of prompting for the user, wherein small variations in wording cause inconsistent results, causing the LLM to hallucinate. The review synthesizes five core techniques in prompting: Manual Prompt Crafting, which emphasizes structural clarity and human control, and Chain-of-Thought (CoT) Prompting, which instructs users to guide the model through a logical "step-by-step" reasoning process. Soft Prompt Tuning, which supports reuse and generalization while retaining a degree of customization. Automated Prompt Generation demonstrates its capacity to generate thousands of prompts relevant to modular task frameworks. However, this scalability has tradeoffs, and at the cost of decreased adaptability and interpretability. Lastly, the study clarifies that Retrieval-Augmented Generation (RAG) is not a standalone technique but a technique that must be complemented by integrating external knowledge retrieval to amplify the effectiveness of prompting. In conclusion, assessing prompt engineering in software engineering is still a dynamic and changing task. Automated evaluation

can be done on a large scale and can be repeated, but it lacks complexity and the ability to be understood for a given task. Human-centric evaluation is more informative, but it lacks scalability and variability. There is an emerging trend among the 42 research studies analyzed that hybrid evaluation methods need to be adopted. Innovation and standardization of metrics, as well as alignment with the domain, are required for future research to adequately capture the efficacy of engineering methods. Prompt engineering can only mature into a sound paradigm for software engineering applications with a proper and context-aware assessment.

2.2.6. Safety Alignment of Large Language Models via Contrasting Safe and Harmful Distributions

Zhang et al. (2025) synthesized existing research on Automatic Prompt Optimization and proposed a comprehensive taxonomy covering core components, including seed prompt selection, evaluation and feedback mechanisms, candidate generation strategies, and selection policies. The review identified heuristic-based approaches, including Genetic Algorithm methods, as a major class of prompt optimization techniques for managing the combinatorial search space of discrete prompts. Across surveyed systems, a consistent trend emerged: optimization effectiveness depends largely on how fitness signals are defined and how candidate prompts are generated and selected. The survey also highlighted that genetic and heuristic approaches are favored for their flexibility and interpretability, particularly in black-box model settings. Based on these findings, the authors recommend designing prompt optimization pipelines with explicit fitness functions, structured evolutionary

operators, and iterative evaluation loops to ensure systematic and reproducible improvements rather than ad-hoc manual refinement.

2.2.7. A Systematic Survey of Prompt Engineering in Large Language Models: Techniques and Applications

According to Sahoo et al. (2024), there is an absence of systematic organization and application-centric understanding of the rapidly expanding set of prompting methods for large language models and vision–language models, which complicates method selection and evaluation across downstream tasks. To address this gap, the authors applied a systematic survey methodology that analyzed over 29 prompting techniques and categorized them by application area, detailing for each technique its prompting approach, target applications, associated model types, and datasets. Afterwards, they consolidated these elements into a taxonomy diagram and summary table that also surfaces strengths and limitations. Their survey characterized prompt engineering as a parameter-efficient mechanism for eliciting task-relevant behavior from pre-trained models across domains such as question answering and commonsense reasoning. However, they also find that there are still significant challenges that persist across technique families, particularly bias, factual inaccuracies, and interpretability gaps. that require continued research attention. Hence, they recommend emerging directions such as meta-learning and hybrid prompting architectures, alongside sustained ethical considerations to guide responsible system development and deployment as prompt engineering becomes increasingly embedded in real-world AI use.

2.2.8. Automatic Prompt Optimization

According to Wolfe (2024), LLM performance can be highly sensitive to prompt wording and structure, yet prompts are commonly improved through manual, trial-and-error iteration that is difficult to systematize because prompts are discrete and many models are accessed via APIs without gradient information. Hence, they provide an overview that distinguishes gradient-based from gradient-free strategies and explains how practical automatic prompt optimization pipelines rely on iterative proposal-and-evaluation loops and on evolutionary-style search procedures, with attention to evaluation design such as training and validation splits, scorer choice, and the use of optimization trajectories to guide successive revisions. Their results show that automated techniques can meaningfully reduce human effort and can sometimes discover prompts that outperform handcrafted baselines. However, they also highlighted limitations that keep such methods assistive rather than fully autonomous, including dependence on a human-provided starting prompt and a tendency to optimize low-level prompt wording while leaving higher-level system design choices. Therefore, they recommended future research that couples prompt optimization with broader LLM-system orchestration frameworks and explores more capable reasoning models in the optimizer role to improve robustness and reduce remaining reliance on manual prompt iteration.

2.2.9. PROMPT ENGINEERING FOR CHATGPT A QUICK GUIDE TO TECHNIQUES, TIPS, AND BEST PRACTICES

Ekin (2023) made a comprehensive analysis and a tutorial on how to properly do prompt engineering and a “tips and tricks” on what to do when facing the mighty

LLM. The study is built on these five pillars. First, prompts should be clear and have specific instructions, which require users to use accurate instructions and detailed descriptions to ensure the LLM understands the exact intent behind a query. Second, prompts should use explicit constraints, such as defining a specific professional tone, limiting word counts, or requesting a particular file format, because this serves as a ground for the model and prevents irrelevant content generation. Third, the user is urged to start experimenting with context and examples. This means providing examples that illustrate the response's style and background details, or a specific "persona" that puts the response in context. Fourth, the researcher outlines the application of System 1 and System 2 questions, distinguishing between "System 1" questions, which seek rapid intuitive responses, and "System 2" questions, which require more in-depth logical reasoning from the model. Finally, it is stressed that controlling output verbosity is a technical requirement, where users can control the length and detail of the response by using direct commands. Additionally, the researchers offered a guide on the best practices for prompting. It includes iterative testing and refining, balancing user intent and model creativity, harnessing external resources and APIs, and ensuring ethical usage and avoiding biases. The researchers suggest developing more sophisticated methods for prompting. Development of stronger models in terms of holding prompt logic in longer interactions. The goal of optimization is to create highly adaptive, domain-specific AI behaviors.

2.2.10. A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT

White et al. (2023) present a list of prompt patterns that are reusable techniques to address typical issues in working with Large Language Models (LLMs) such as ChatGPT, Claude, and Gemini. Prompt engineering, according to the authors, is a form of natural language programming that aims to maximize interaction dynamics, input, and output. There are 13 prompt patterns available in the list, and these qualify under 6 core categories. The first category is Context Control, which deals with managing the data that the LLM uses to ensure that the responses are accurate and logical. The second category is input semantic patterns, which define how the LLM will interpret the user input and then process it into formats that can be used. Third, Output Customization, this category restricts or tailors the types, formats, and layouts of the LLM's output. The fourth category is Error Identification, which emphasizes error recognition and correction in the content that is produced by the LLM. The fifth category is Prompt Improvement. These categories are designed to improve the quality of the output that is generated as well as the input that is provided to the LLM. Lastly, Interaction, these patterns control the dynamics and flow of the conversation between the model and the user. These six core categorical patterns provide a reusable solution when facing a problem in interacting with LLMs to perform a specific task. The authors also recommend combining the prompting patterns under these categories to yield a stronger and desirable output.

2.2.11. ChatGPT Prompt Patterns for Improving Code Quality, Refactoring, Requirements Elicitation, and Software Design

A similar study by White, et al. (2023) was made for prompt engineering, but specifically for the model ChatGPT and for coding tasks only. The researchers presented 14 prompt patterns categorized into four core domains: Requirements Elicitation, System Design and Simulation, Code Quality, and Refactoring. To improve the technical precision of Large Language Model (LLM) outputs and resolve common development bottlenecks. The first is the Requirements Simulator, which allows stakeholders to interactively explore if system functionality is captured properly. Second, Specification Disambiguation is used to find points of weakness or ambiguity in requirements before they reach developers. Third, the API Generator produces an API specification, such as a REST API specifications from natural language descriptions. Fourth, the API Simulator enables developers to validate the API specification by simulating responses using the generated specification. Fourth, the API Simulator enables the developer to test the API specification. Fifth, Principled Code facilitates the creation of code that adheres to design principles such as SOLID without requiring the specification of all the rules. Sixth, Intermediate Abstraction forces the LLM to introduce layers between the business logic and third-party libraries to prevent tight coupling. Finally, Data-guided Refactoring enables high-level code refactoring by providing the LLM with new data samples to control the refactoring process. In the end, the researchers revealed that the capabilities of the LLMs are underappreciated and their capabilities are not fully utilized. They suggested that a significant amount of human intervention and

expertise is necessary to fully understand the abilities of an LLM and to produce competent outputs. Thus, they strongly suggest the possibility of combining these prompting techniques to maximize the potential of the LLM.

2.3. Synthesis of Local and Foreign Literatures

Local studies highlight a rapid but largely unmanaged adoption of artificial intelligence tools in the Philippines. The country ranks sixth globally in ChatGPT usage, with adoption rates nearly double the global average (Mateo, 2025). This high usage is reflected in education, where many students and teachers actively rely on generative AI for academic assignments, translations, and learning materials (Martinez, 2025). While AI use is also expanding across other sectors, local researchers emphasize that this growth is occurring without unified governance standards, formal evaluation frameworks, or technical mechanisms to ensure output quality (Santos & Jocson, 2024). In contrast, foreign literature shifts the discussion from adoption to control mechanisms, emphasizing prompt engineering as a structured method for managing AI outputs through defined interaction patterns (White et al., 2023), constraint specification, and systematic refinement strategies (Sahoo et al., 2024).

The combined insights from these literatures strongly support the Garbage In, Garbage Out (GIGO) principle, which states that an AI model's accuracy is directly dependent on input quality (Shelf, 2025). To prevent hallucinated, irrelevant, or inconsistent responses, prompts must demonstrate clarity, specificity, and proper context (Ekin, 2023). This perspective parallels requirements engineering principles, where structured elicitation, defined objectives, and explicit constraints are necessary to accurately capture user intent

before system execution (White et al., 2023). Moreover, the literature consistently identifies the problem of "prompt brittleness," wherein minor, meaning-preserving variations in phrasing produce significantly different outputs (Syahputri et al., 2025; Schulhoff et al., 2025). This issue reveals structural instability in prompt-model interactions and underscores the importance of semantic preservation and controlled evaluation environments (Schulhoff et al., 2025). Additionally, structured prompting not only improves output quality but also serves as an educational mechanism for users to develop better prompting habits, ultimately reducing hallucinations and misuse of AI systems (Liu et al., 2025; Ekin, 2023).

Despite the availability of numerous prompting techniques and optimization tools, a critical research gap persists in systematic evaluation and optimization. Automated prompt optimization approaches reduce manual effort but still depend on initial human prompts and defined objective functions (Wolfe, 2024; Syahputri et al., 2025). Moreover, benchmarking practices remain inconsistent, with limited multi-dimensional metrics capable of capturing clarity, specificity, semantic stability, and output reliability across domains (Schulhoff et al., 2025). By incorporating multi-criteria decision-making (MCDM) attributes such as prompt clarity, specificity, and semantic preservation, researchers can develop more systematic and consistent evaluation frameworks. Addressing these gaps may support the shift from unmanaged AI usage toward a more structured, reliable, and standardized practice.

REVIEW OF RELATED STUDIES

This presents a review of the relevant literature related to the present research. It includes both foreign and local literature that provide theoretical and empirical support to the variables investigated. The synthesis highlights gaps in previous research, which justify the conduct of the present study.

2.4. Local Related Studies

2.4.1. Between Innovation and Tradition: A Narrative Inquiry of Students' and Teachers' Experiences with ChatGPT in Philippine Higher Education

Espartinez (2025) employed a narrative inquiry approach, using Clandinin and Connelly's three-dimensional framework, to examine the lived experiences of 18 participants, comprising students and faculty members from Philippine higher education institutions in both urban and rural contexts. The findings reveal widespread but largely unregulated use of ChatGPT, described as "silent acceptance," alongside strong ethical concerns and pedagogical tensions. A key Philippine trend highlighted is the persistent urban–rural digital divide, with only approximately 40% of rural areas having reliable internet access, compared with approximately 80% in urban areas. Participants reported a shift from initial dependence on ChatGPT toward more reflective use; however, the study underscores the absence of structured frameworks to guide effective and ethical interactions with AI. The research does not provide technical mechanisms for optimizing prompt quality. Furthermore, the author

recommends developing context-sensitive AI policies, faculty training initiatives, and equitable access strategies, including low-resource or offline AI solutions, to ensure responsible and inclusive adoption in higher education.

2.4.2. Algorithmic Window Shopping: The Impact of Generative AI on the Purchase Decisions of Filipino Gen Z Consumers

Moreno et al. (2025) examined how exposure to generative AI influences purchase intentions among Filipino Generation Z consumers in online shopping contexts. Using a quantitative survey design with purposive sampling, the study analyzed responses from 208 Gen Z participants and tested relationships among AI exposure, attitudes, perceived accuracy, trust, ease of use, and purchase intention using structural equation modeling. The findings revealed that AI exposure, perceived accuracy, and positive attitudes toward AI significantly increased intention to use AI-driven systems, which in turn strongly influenced trust and perceived ease of use, ultimately affecting purchase intention. Trust and ease of use were identified as mediating factors in the relationship between AI utilization and consumer decision-making. Although the study is situated in a commercial rather than academic or prompt engineering context, it reinforces a consistent pattern: perceived output quality and user trust shape adoption and reliance on AI systems. The authors recommend that organizations ensure transparency, system reliability, and user safeguards to strengthen trust and sustain long-term adoption of generative AI technologies.

2.4.3. Trust in Generative AI among Students

Amoozadeh et al. (2024) explored students' trust in generative AI systems and their relationship with perceived motivation and confidence in programming-related tasks. The researchers employed an exploratory survey design with 253 university students, using a trust-in-automation scale and descriptive statistics, correlation analysis, and qualitative coding of open-ended responses. The study found that 47 percent of respondents expressed trust in generative AI, while 36 percent were neutral and 16 percent reported distrust. Higher trust levels were associated with increased motivation and confidence, particularly among first-generation students, suggesting that perceived reliability of AI outputs plays a significant role in user engagement. Qualitative findings revealed that students remained cautious, frequently emphasizing the need for verification and human oversight due to potential inaccuracies and biases in AI-generated responses. The authors recommend that educators and developers promote calibrated trust by enhancing transparency and enabling users to critically assess AI-generated outputs rather than relying on them unquestioningly.

2.4.4. The role of ChatGPT on academic research: perspectives from Filipino students across diverse educational levels

Cabuquin et al. (2024) conducted a descriptive cross-sectional study involving 316 Filipino students from secondary and tertiary institutions in Eastern Visayas to examine perceptions, benefits, and challenges associated with the use of ChatGPT in academic research. The findings indicate that ChatGPT is perceived as moderately useful and moderately influential on students' academic performance, particularly in tasks such as idea generation, paraphrasing, writing mechanics, and accessing

supplementary resources. Despite these benefits, students frequently encountered issues, including redundant, inaccurate, and inadequate responses, as well as concerns regarding plagiarism and accessibility. Moreover, Filipino students are increasingly relying on ChatGPT despite awareness of its limitations. However, the study does not analyze how prompt formulation or semantic structure affects the quality of AI-generated outputs, leaving a gap in understanding how structured or optimized prompts could address the reported inaccuracies. The authors recommend that academic institutions establish clear policies and guidelines to promote responsible and effective use of ChatGPT, supported by instructional interventions that maximize benefits while mitigating ethical risks.

2.4.5. Assessing AI Adoption: Investigating Variances in AI Utilization across Student Year Levels in Far Eastern University-Manila, Philippines

Dela Rosa et al. (2024) investigated the adoption of artificial intelligence among undergraduate students at a private university in Manila using a quantitative survey design with 40 respondents across multiple year levels. The study found that motivations for AI usage varied by academic standing, including peer influence, academic performance pressure, time efficiency, and error reduction. Statistical analysis revealed no significant difference in the likelihood of AI use across year levels, indicating that AI tools are already normalized within students' academic workflows regardless of experience. Despite its widespread use, the authors identified concerns regarding overreliance on AI and potential threats to academic integrity. While the study provides insight into behavioral and motivational aspects of AI use among Filipino students, it does not address technical mechanisms for improving

interaction quality between users and AI systems. Therefore, the authors recommend stronger institutional policies, AI literacy initiatives, and monitoring mechanisms to promote responsible usage.

2.4.6. Empowering Creative Education: Applying Chat GPT for Enhancing Student Engagement in Senior Teacher-Driven Instructional Design in the Philippines

Hatmanto et al. (2024) conducted a qualitative study using semi-structured interviews with 10 senior educators in Mindanao and applied content analysis to examine strategies for integrating ChatGPT into instructional design to enhance student engagement. The findings reveal a range of teacher attitudes toward ChatGPT, from enthusiasm to skepticism, with effective practices including tailored prompts, structured inquiry, and guided modes of expression. The study identified the influential role of experienced educators in mediating the use of AI to support creativity and engagement in the classroom. However, prompt design is treated as an intuitive teaching practice rather than an evaluated process. The authors recommend further research into structured frameworks for AI-supported pedagogy and deeper exploration of prompt-guided instructional strategies to improve learning outcomes.

2.4.7. Readiness for AI adoption of Philippine business and industry: The government's role in fostering innovation and AI-driven industrial development

Quimba et al. (2024) examined artificial intelligence readiness and adoption across Philippine industries using national datasets and policy-oriented analysis, including data from the Philippine Statistics Authority’s Survey on ICT. Their findings indicate

that while basic digital infrastructure is widespread, over 90% of establishments possess computers and more than 80% have internet access, advanced digital engagement remains limited: only 21.7% maintain websites, and 31.2% participate in e-commerce. Notably, only 14.9% of firms reported using AI technologies, with an overall national AI adoption rate of approximately 3.02%, concentrated mainly in ICT- and BPO-related sectors and significantly lower in agriculture and small enterprises. The study highlights a pronounced digital divide influenced by firm size, sector, and geographic location, suggesting that access to technology does not necessarily translate into effective AI utilization. The researchers recommend strong government intervention through infrastructure investment, skills development, and ecosystem coordination to promote inclusive and sustainable AI adoption nationwide.

2.4.8. Converging perspectives: Assessing AI readiness and utilization in Philippine higher education

Asirit and Hua (2023) employed a mixed-methods research design to assess artificial intelligence readiness and utilization among college students in Philippine higher education institutions. Quantitative survey results from 250 respondents revealed a moderate level of AI awareness: approximately 79% of students reported some familiarity with AI technologies, whereas about 21% reported low awareness, indicating a significant AI literacy gap. Qualitative findings further showed that students generally perceive AI positively as a tool for efficiency and academic support, although concerns regarding academic integrity, data privacy, and employment displacement were consistently expressed. The study also found that AI readiness varies across academic disciplines, year levels, and exposure to AI-related

learning experiences. While the research demonstrates increasing engagement with AI among Filipino students. Additionally, the authors recommend targeted AI education, curriculum integration, and ethical guidance to ensure the responsible and effective use of AI in higher education.

2.4.9. Artificial Intelligence for Sustainability: Evidence from select Small and Medium Enterprises in the Philippines

Hernandez et al. (2023) examined how small and medium enterprises in the Philippines employ artificial intelligence to support sustainability initiatives. Using a qualitative research design, the authors conducted in-depth, semi-structured interviews with decision-makers from 16 SMEs that had implemented AI-related projects and analyzed the data thematically. The study highlighted the economic importance of SMEs, which account for approximately 99.5 percent of enterprises and around 35.7 percent of national value added, while revealing that only a limited number have adopted AI for sustainability-related purposes. Reported AI applications included customer-facing chatbots, sales and demand forecasting tools, AI-driven product recommendation systems, and automated issue tracking platforms, with most initiatives described as early-stage and incremental rather than fully integrated. The findings also identified recurring constraints, including limited infrastructure, insufficient data readiness, regulatory uncertainty, privacy concerns, and the absence of clear organizational AI strategies, collectively hindering broader adoption. The authors recommend strengthening institutional support, improving infrastructure and governance frameworks, and developing clearer strategic directions to enable SMEs to scale AI initiatives in a sustainable and responsible manner.

2.5. Foreign Related Studies

2.5.1. Rethinking Prompt Optimizers: From Prompt Merits to Optimization

According to Zhu et al. (2026), there is a prevailing automatic optimization that is produced by advanced LLMS and is often wordy and instruction-heavy. This creates limited "downward compatibility" that can overwhelm lightweight inference models and obscure interpretability due to implicit optimization, while online optimization paradigms introduce cost and privacy concerns. To address this, they adopted an explicit-design methodology that first operationalized model-agnostic prompt merits including clarity, precision, and concise reasoning guidance, and then built a merit-guided prompt optimization preference dataset (POP) of 40,000 comparisons by having a lightweight LLM generate prompt candidates and using a stronger evaluator to judge prompt and response quality; these data supported training MePO, a locally deployable prompt optimizer via preference learning. Moreover, the results show that merit-guided optimized prompts were preferred in 92.3% of pairwise comparisons and that MePO improved response quality across both large-scale and lightweight inference models. However, the authors also documented that MePO remains a one-shot optimizer lacking interactive feedback and that further gains may require stronger model adaptation under architectural mismatch. Hence, they recommended future work on integrating user/inference feedback loops and training model-adaptive optimizers to improve robustness in real-world settings.

2.5.2. A Comprehensive Approach to Instruction Tuning for Qwen2.5: Data Selection, Domain Interaction, and Training Protocols

Gu et al. (2025) conducted controlled experiments on Qwen2.5-3B and 7B models to evaluate the effects of data selection, domain interaction, and instruction-tuning protocols. Their results showed that optimized dataset composition improved average benchmark performance by approximately 8–12% across reasoning and instruction-following tasks compared to baseline fine-tuning setups. The study further observed improved generalization when heterogeneous domains were strategically mixed during training. However, the experiments primarily focused on dataset and training protocol variations and did not systematically examine prompt-level structural changes. The authors recommend further investigation into how input design strategies may interact with training protocols to influence performance stability.

2.5.3. AgentReport: A Multi-Agent LLM Approach for Automated and Reproducible Bug Report Generation

Choi and Yang (2025) implemented QLoRA-based fine-tuning on Qwen2.5-7B-Instruct under resource-constrained conditions to evaluate parameter-efficient adaptation. Their findings indicate that 4-bit quantized fine-tuning achieved performance within 1–2% of full-precision fine-tuning, while significantly reducing memory consumption and hardware requirements. The study highlights the feasibility of efficient transformer adaptation without substantial loss in accuracy. However, evaluation focused mainly on task accuracy and computational efficiency, with limited assessment of prompt variability, robustness across input formulations, or token usage optimization. The authors suggest exploring additional optimization dimensions beyond parameter adaptation.

2.5.4. Qwen2.5 Technical Report

The Qwen Team (2025) introduced Qwen2.5 as an enhanced series of transformer-based causal language models trained on expanded large-scale pre-training data and refined through supervised fine-tuning and multi-stage reinforcement learning. The models were evaluated across benchmarks covering language understanding, reasoning, mathematics, coding, and human preference alignment, where Qwen2.5-72B-Instruct exhibited top-tier cross-domain performance with parameter-efficient near-SOTA competitiveness. Despite being approximately five times smaller, the model remained competitive with larger state-of-the-art open-weight counterparts, highlighting the effectiveness of its training and alignment

pipeline. The study suggests that strategic data scaling and structured post-training optimization can yield highly efficient models that rival substantially larger architectures.

2.5.5. Advances in neural text generation: A systematic review (2022-2024)

Slobodianiuk & Semerikov (2025) conducted a systematic literature review that analyzed 43 articles from the Scopus database to summarize the methods, architectures, and evaluation trends in neural text generation from 2022 to 2024. It was stated that metrics such as BLEU, ROUGE, and Human Evaluation remain popular, while new metrics like BERTScore are showing interest. The study highlighted that the field of neural text generation is increasingly expanding and opening up possible explorations for further related studies.

2.5.6. RST-LoRA: A Discourse-Aware Low-Rank Adaptation for Long Document Summarization

Liu and Demberg (2024) proposed a discourse-aware LoRA variant for long document summarization, integrating rhetorical structure signals into adapter layers. Their results showed 3–9% improvements in human preference evaluations compared to baseline LoRA and full fine-tuning approaches, while updating less than 0.5% of model parameters. Automatic evaluation metrics such as ROUGE also reflected consistent gains. The study focused on discourse-aware adaptation rather than prompt formulation variability. The authors suggest that additional structural components and evaluation criteria may further enhance summarization performance.

2.5.7. LoRA Learns Less and Forgets Less

Biderman et al. (2024) compared LoRA and full fine-tuning across programming and mathematics benchmarks to evaluate learning capacity and forgetting behavior. Their results showed that full fine-tuning outperformed LoRA by 8–15% on target-domain tasks, while LoRA better preserved out-of-domain capabilities, demonstrating improved robustness and reduced catastrophic forgetting. They also found that full fine-tuning introduced parameter perturbations with rank values 10–100× larger than typical LoRA configurations. The study focused on parameter-level trade-offs and did not analyze prompt-level optimization strategies. The authors recommend further examination of adaptation mechanisms that balance accuracy and robustness.

2.5.8. GoRA: Gradient-Driven Adaptive Low-Rank Adaptation

He et al. (2025) proposed Gradient-Driven Adaptive Low-Rank Adaptation (GoRA), which dynamically adjusts adapter rank based on gradient importance. Experimental results showed improvements of 5.88 points on GLUE and 5.13 points on GSM8K compared to standard LoRA configurations. The method demonstrated that adaptive rank allocation enhances fine-tuning performance under constrained parameter budgets. The study emphasized task accuracy improvements and parameter efficiency but did not include structured analysis of prompt design or multi-dimensional evaluation criteria. The authors recommend exploring broader optimization frameworks that incorporate additional performance indicators.

2.5.9. Prompt's Evolution for Language Model-Driven Data Generation

According to Hidalgo et al. (2025), there is limited or imbalanced real-world data by evolving prompts that instruct an LLM to produce diverse and semantically relevant text, effectively augmenting training datasets for stream processing systems. To address this gap, they developed a novel genetic algorithm-based prompt optimization framework to generate synthetic textual data with the aid of large language models, targeting applications in data-scarce domains like disaster management. The proposed system combined an LLM, which is kept as a fixed black-box text generator, with an evolutionary strategy, guided by a fitness function measuring the similarity of LLM-generated outputs to a reference dataset like authentic disaster-related social media posts, while also encouraging output diversity. Moreover, the results showed that this evolutionary prompt approach significantly improved the semantic fidelity of the generated data to the target domain, while maintaining a high level of diversity in the synthetic texts. The outcome is a generalizable framework that can automatically produce high-quality data for downstream models, which demonstrates a particular value when direct fine-tuning of an LLM is infeasible. However, Hidalgo et al. noted that the quality of generated data correlates with the complexity of the target text and that careful calibration of GA parameters is needed for efficiency. Hence, they recommend future enhancements, including designing more efficient fitness evaluation functions to reduce the computational cost of LLM calls, using sentence-level evaluation to speed up convergence, and integrating reinforcement learning techniques to dynamically guide the evolution process for an even better balance between exploration and exploitation in prompt optimization.

2.5.10. Characterizing Fitness Landscape Structures in Prompt Engineering

According to Hintze (2025), there is an unexamined question of what the "shape" of the prompt optimization landscape looks like, as most prompt tuning approaches blindly apply search algorithms without understanding the underlying topology. To address this, using autocorrelation analysis in a semantic embedding space, the researcher characterized the landscape by generating prompts for an error-detection task under two conditions: one with a systematic enumeration of prompt variants and another promoting novelty and diversity among prompts. Their analysis revealed that systematically generated prompts produced a smooth landscape with gradually decaying similarity as prompts differed more, whereas the diversity-driven approach led to a rugged, hierarchical landscape with irregular correlation patterns. Furthermore, across ten different error categories, the degree of landscape ruggedness varied, indicating that some prompt optimization problems are intrinsically harder or more locally deceptive than others. These findings provide insight into why certain optimization methods succeed or fail for prompt engineering; for instance, a smooth landscape might be efficiently traversed by simple hill-climbing, whereas a rugged landscape may require more robust global search or diversity preservation. Hence, recommending that prompt engineers assess landscape features before choosing an optimization technique and suggesting that future work develop adaptive algorithms that account for the identified landscape structures to improve prompt optimization outcomes.

2.5.11. An LLM-Based Genetic Algorithm for Prompt Engineering

According to Loss and Dhuvad (2025), prompt quality often depends on specialized prompt engineering skill and domain-specific expertise, which can increase costs and lead to suboptimal performance when such expertise is absent. Hence, they proposed an LLM-based genetic algorithm for prompt engineering that customizes standard GA machinery to operate on textual individuals, using LLM-guided mutation and crossover operators to iteratively generate and refine candidate prompts and employing LLM-based output judging as the fitness signal to guide selection across generations. Based on their experiments spanning four public datasets and multiple modern LLMs from different vendors, GA-derived prompts consistently yield solutions superior in both accuracy and efficiency compared with prompts produced by technically skilled engineers lacking domain-specific knowledge. However, their work also implied a future research need to further validate and strengthen evaluator reliability and to extend GA-driven prompt evolution to broader tasks and deployment settings so that automated optimization tools can reduce human trial-and-error without sacrificing robustness.

2.5.12. Constructing Differential Evolution via LLM Prompt Chaining

According to Papaj et al. (2025), LLMs, when given properly structured prompts, can generate sophisticated optimization algorithms without human-coded logic. Hence, they explored an innovative approach to algorithm design by using a large language model (LLM) to construct a differential evolution (DE) algorithm via prompt chaining. They guided an LLM through a sequence of prompts to create "DEuLLM," a complete DE algorithm defined by the LLM's outputs, including

population initialization, mutation, crossover, and selection steps. Moreover, they treated the LLM as a black-box designer: iterative prompt refinements ensured the generated algorithm met the typical properties of DE. Experimental evaluation on standard numerical optimization benchmarks showed that the LLM-designed DE was functional and could find reasonable solutions, though with some performance gap compared to a hand-crafted DE variant. Hence, they note a potential of prompt engineering to automate algorithm development and suggest that future research could improve such LLM-generated algorithms by refining prompt strategies, incorporating human-in-the-loop feedback, and ensuring greater originality and efficiency in the evolved solutions.

2.5.13. GAAPO: genetic algorithmic applied to prompt optimization

According to Sécheresse et al. (2025), manual prompt engineering is time-consuming and prone to human error, and prior genetic-algorithm prompt optimizers tend to operate within narrow mutation/crossover paradigms that lack extensibility and modularity for integrating broader prompt transformation strategies. Hence, they introduced GAAPO, a hybrid evolutionary framework that incorporates multiple specialized prompt-generation strategies within each generation and evaluates candidates using alternative selection schemes such as complete evaluation, bandit-based selection, and successive halving. The framework was tested on ETHOS multilabel hate-speech classification and knowledge/reasoning benchmarks such as MMLU-Pro and GPQA, with comparisons against established prompt optimizers and across different LLM prompt generators. Their results showed that GAAPO outperformed baseline methods on tasks with meaningful optimization headroom,

increasing ETHOS validation performance from 0.28 to 0.46. This demonstrates that the selection strategy substantially affects stable optimization trajectory than more volatile alternatives. However, they still acknowledged limitations, including constrained statistical validation due to computational costs, task-dependent variability, and the persistent restriction of prompt evaluation overhead, while also noting that fairness and bias evolution were outside its scope. Therefore, they recommended expanding statistical runs, developing adaptive strategy-weighting mechanisms, improving generalization metrics, reducing computational overhead, extending evaluation to broader NLP task families, and auditing ethical risks such as potential bias amplification during prompt evolution.

2.5.14. Pareto Prompt Optimization

Zhao et al. (2025) proposed Pareto Prompt Optimization, a multi-objective prompt optimization approach designed to handle scenarios in which prompt quality must satisfy multiple competing criteria simultaneously. The method applies preference-based reinforcement learning guided by Pareto dominance, comparing pairs of prompts to determine whether one dominates the other across multiple objectives, and updates the optimization policy using dominance-preference losses without requiring predefined scalar weights. To encourage diversity and avoid premature convergence, the approach incorporates a non-dominated loss that promotes exploration across the Pareto front. Experimental results showed that ParetoPrompt consistently outperformed baseline multi-objective optimization methods that rely on fixed objective aggregation, and remained robust even when evaluation metrics differed between training and testing conditions. However, the

authors noted that as the number of objectives increases, dominant prompt pairs become less frequent, thereby reducing optimization efficiency and increasing sampling complexity. They recommend dominance-driven preference learning as a robust strategy for multi-objective prompt optimization while emphasizing the need for careful loss design and sampling strategies to maintain efficiency in many-objective settings.

2.5.15. A Comprehensive Survey of Direct Preference Optimization: Datasets, Theories, Variants, and Applications

Xiao et al. (2025) provides a structured synthesis of the theoretical foundations, dataset constructions, algorithmic variants, and practical applications of DPO in large language model alignment. The study explains how DPO reformulates reinforcement learning from human feedback into a supervised objective that directly optimizes pairwise preference data without requiring an explicit reward model or policy optimization loop. Through systematic categorization of DPO variants and applications across instruction tuning, safety alignment, and domain-specific tasks, the authors highlight its computational efficiency and training stability compared to PPO-based RLHF approaches. However, the survey emphasizes that DPO's effectiveness depends heavily on the quality and consistency of human preference datasets, and that robustness to noisy labels or distributional shifts remains underexplored. The authors recommend developing standardized benchmarking protocols, improving dataset construction methodologies, and extending DPO toward multi-objective and more robust alignment settings to strengthen its theoretical and practical reliability.

2.5.16. Aligning CodeLLMs with Direct Preference Optimization

Miao et al. (2024) investigate the effectiveness of applying DPO to code generation. The authors construct pairwise human preference datasets for programming tasks and apply DPO as a post-supervised fine-tuning alignment stage, comparing its performance with supervised fine-tuning and reinforcement learning-based methods. Empirical results demonstrate improvements in functional correctness and instruction-following behavior across standard coding benchmarks, suggesting that DPO can effectively refine structured output generation without the instability associated with reinforcement learning. Despite these gains, the study acknowledges limitations such as the high cost of curating expert-annotated code preference datasets and the potential benchmark-specific improvements that may not generalize across diverse programming environments. The authors recommend expanding multilingual and multi-language code datasets, integrating heuristic filtering mechanisms before preference optimization, and conducting broader evaluations to assess robustness across varying code complexities.

2.5.17. Meta-Prompting: Enhancing Language Models with Task-Agnostic Scaffolding

Suzgun and Kalai (2024) introduced a meta-prompting framework that coordinates multiple expert-style prompts through a conductor mechanism for reasoning tasks. Their experiments reported approximately 15–17% higher accuracy compared to standard scaffolding and multipersona prompting methods across reasoning

benchmarks. While these results demonstrate substantial performance gains, the approach requires multiple sequential model invocations, increasing computational cost. Evaluation was primarily based on task accuracy metrics. The authors acknowledge the added computational overhead and recommend investigating efficiency-aware optimization strategies.

2.5.18. Black-Box Prompt Optimization: Aligning Large Language Models without Model Training

Cheng et al. (2024) proposed Black-Box Prompt Optimization (BPO) as an alignment strategy that improves large language model behavior by optimizing user prompts rather than modifying model parameters, making the approach applicable to closed, API-based models. The study employed a criticize-and-optimize procedure, using preference-style datasets to generate paired original and optimized prompts, enabling the training of a sequence-to-sequence prompt optimizer that rewrites instructions based on feedback. Experimental results demonstrated that BPO-optimized prompts consistently outperformed original prompts across multiple language model families, with reported win-rate improvements of up to 22 percentage points for GPT-3.5-based models and over 10 percentage points for GPT-4, indicating that structured prompt rewriting can significantly improve alignment and instruction-following. The authors also observed diminishing returns after multiple optimization rounds, highlighting the need for principled stopping criteria and evaluation control. They recommend further research into robust evaluation signals and controlled optimization strategies to prevent over-optimization while maintaining performance gains

2.5.19. EVOPROMPT: Connecting LLMS with Evolutionary Algorithms Yields Powerful Prompt Optimizers

According to Guo et al. (2024), there is a gap that effective prompt engineering for large language models typically requires substantial human effort, and that standard evolutionary operators are poorly suited to discrete prompts because token-level edits can undermine coherence and readability. To address this, they introduced EvoPrompt, which formulates prompt search as a black-box evolutionary optimization process that maintains a population of prompts, uses an LLM to realize mutation and crossover-like transformations in natural language, and updates the population via selection based on development-set performance without requiring access to model gradients or internal parameters. To evaluate generality, they tested the framework on 31 datasets spanning language understanding, text generation, and BIG-Bench Hard tasks, and it was applied across both closed-source and open-source LLMs, including GPT-3.5 and Alpaca. Results indicated that EvoPrompt consistently outperformed manually engineered prompts and prior automatic prompt-generation baselines, with reported gains reaching up to 25% on BBH, supporting the claim that coupling LLM generation with evolutionary search yields complementary strengths. Hence, they suggest that the same LLM–EA synergy could be extended to other optimization settings that require coherent natural-language solutions, and recommend exploring more advanced evolutionary strategies and broader task domains while managing the computational cost of repeated LLM queries.

2.5.20. Evolutionary Computation for the Design and Enrichment of General-Purpose Artificial Intelligence Systems: Survey and Prospects

According to Poyatos et al. (2024), configuring and adapting models as an AI system expand to handle multiple diverse tasks is a key challenge and something traditional single-task design struggles with. To address this gap, they examined how various evolutionary techniques, such as genetic algorithms, genetic programming, neuroevolution, etc., have been used to automatically design, optimize, or adapt machine learning models. They highlighted notable successes of evolutionary computations (EC), including evolving neural network architectures and hyperparameters, which map well to the adaptability requirements of GPAIS. However, they also discuss open challenges in leveraging EC for GPAIS, including the computational complexity of evolving large-scale AI components and the need to ensure reliability across tasks. Hence, this outlined several future research directions, recommending more research into hybrid approaches that combine EC with deep learning for self-configuring AI, and calling for the development of evolutionary strategies specifically tailored to manage the complexity and dynamic nature of general-purpose AI systems.

2.5.21. Teach Better or Show Smarter? On Instructions and Exemplars in Automatic Prompt Optimization

Wan et al. (2024) investigated how different components of prompts, specifically task instructions and exemplars, contribute to the effectiveness of automatic prompt optimization. Using empirical evaluations across several tasks and model settings, the study compared optimization strategies that refine instructions alone, exemplars

alone, and both components jointly. The findings revealed that jointly optimizing instructions and exemplars consistently produced the strongest performance improvements, with average scores increasing from 60.3 for the original prompt to 76.1 under joint optimization, outperforming strategies that optimized either component in isolation. The study further observed that the relative contributions of instructions and exemplars varied across tasks and model contexts, indicating that prompt optimization outcomes are sensitive to structural design choices. Based on these results, the authors recommend prompt optimization frameworks that explicitly account for different prompt components and rely on systematic evaluation rather than ad-hoc manual refinement.

2.5.22. Large Language Models as Optimizers

According to Yang et al. (2024), there is a gap in derivative-free optimization settings where gradients are unavailable and conventional optimizers often require task-specific customization, limiting their practicality, such as in the search for effective prompts. Hence, they introduced Optimization by PROmpting (OPRO), a general-purpose, iterative framework that uses an LLM as the optimizer: each step prompts the model with a natural-language problem description plus a ranked history of previously generated solutions and their objective values, produces new candidate solutions, and then relies on an external evaluator to score and append candidates for the next iteration. Using case studies in linear regression and traveling salesman problems and a primary application to prompt optimization on GSM8K and Big-Bench Hard, OPRO yielded optimized instructions that substantially improved downstream performance, with best prompts surpassing human-designed prompts by

up to 8% on GSM8K and up to 50% on Big-Bench Hard tasks. On the other hand, they noted limitations including risks of prompt overfitting when optimization is driven by small evaluation sets and sensitivity to initialization and search hyperparameters; thereby recommending for future work to incorporate richer feedback beyond aggregate accuracy, and summarize the key features that distinguish between high-quality and low-quality generated prompts in the optimization trajectory.

2.5.23. Do Clinicians Know How to Prompt? The Need for Automatic Prompt Optimization Help in Clinical Note Generation

Yao et al. (2024) examined variability in prompt quality within clinical note generation tasks by comparing generic prompts, prompts modified by medical experts and non-experts, and prompts refined through an automatic prompt optimization framework applied to large language models. The study employed both automatic evaluation metrics, including ROUGE, METEOR, and a medical concept-based factuality score, as well as human preference assessments. Results showed that prompts optimized by the automatic framework consistently outperformed manually modified prompts in content quality and factual coverage, particularly when paired with more advanced language models. The findings also indicated that while expert users could sometimes match optimized prompts through careful manual refinement, non-expert users frequently experienced a decline in quality without algorithmic support. The authors recommend a hybrid workflow in which automatic prompt optimization ensures baseline quality and consistency, followed by domain-expert

refinement for personalization, reducing the cognitive burden associated with prompt engineering in specialized domains.

2.5.24. Automatic Prompt Optimization with Textual Gradients

Ye et al. (2023) proposed an automatic prompt optimization approach using textual gradient feedback generated by language models. Their experiments demonstrated measurable improvements in reasoning benchmarks, including GSM8K, relative to initial prompt configurations. The method iteratively refines prompts using model-generated feedback signals. While performance gains were observed, evaluation focused primarily on accuracy-based metrics. The authors note potential sensitivity to feedback quality and recommend exploring more comprehensive evaluation strategies that consider stability and efficiency factors.

2.5.25. MAPO: Boosting Large Language Model Performance with Model-Adaptive Prompt Optimization

LLMs rose in the new era, but still depend on the quality of the prompts. Chen et al. (2023) developed a model that optimized the standards of a prompt, which they named MAPO (Model-Adaptive Prompt Optimization). The framework begins by establishing a warm-up dataset for prompt optimization. The researchers generated 1,000 candidate prompts that should maintain the same semantic meaning but are expressed differently. These candidates are then ranked to find the “optimal prompt”. After that, the multi-layered process begins. It starts with Supervised Fine-Tuning (SFT), which aligns the target LLM with these optimal prompts. Second, a Reward Model is trained to assign scalar scores to the prompts and is ranked according to the

model's preference. Finally, the system uses Reinforcement Learning (RL), with Proximal Policy Optimization (PPO) and a new approach called Ranking Responses from Model Feedback (RRMF), to further refine the prompts. To stabilize the model and prevent it from losing its general abilities, MAPO uses Joint Learning. The researcher observed that the performance of the system increases during SFT, and the optimized prompts that are yielded are more adaptive in QA and BLOOM models. MAPO is also more adaptive in classification tasks, which shows that it amplifies the performance of various LLMs and shows a preference in different downstream tasks. The study recommends the construction of more fine-grained model adaptive prompts that can adapt to the ever evolving data encountered in real-world environments, and to enhance the applicability to be able to cater to a wider range of linguistic contexts.

2.5.26. Automatic Prompt Optimization with "Gradient Descent" and Beam Search

Pryzant et al. (2023) introduced Prompt Optimization with Textual Gradients (ProTeGi), an automatic prompt optimization approach designed for black-box large language model APIs. The method iteratively evaluates an initial prompt using minibatches of task data, generates natural-language feedback that identifies weaknesses in the prompt, and applies semantic edits in the opposite direction of these identified flaws, while employing beam search and a bandit-style selection mechanism to efficiently manage the candidate search space. Experimental results across multiple natural language processing tasks demonstrated that the optimized prompts improved performance by up to 31 percent over the initial prompts and outperformed prior prompt-optimization baselines by an average margin of 4-8

percent, while also reducing the number of required API calls. Despite these gains, the authors note limitations related to computational cost, sensitivity to evaluation budget, and the relatively small number of benchmark tasks considered. They recommend future work that expands task coverage, explores alternative evaluation metrics, and improves optimization efficiency to make automatic prompt refinement more scalable and broadly applicable.

2.5.27. Direct Preference Optimization: Your Language Model is Secretly a Reward Model

Rafailov et al. (2023) proposed Direct Preference Optimization (DPO) as a reinforcement learning–free alternative to traditional RLHF for aligning language models with human preferences. The method reformulates the standard RLHF objective using the Bradley-Terry preference model and reparameterizes the reward function in terms of the ratio between the policy and a reference model, allowing optimization through a simple binary cross-entropy loss rather than PPO-based reinforcement learning. Empirical evaluations on sentiment control, summarization, and dialogue tasks demonstrate that DPO achieves comparable or superior performance to PPO while being more stable and computationally efficient. However, the study is limited to models up to 6B parameters, relies heavily on GPT-4 as an automated evaluator, and does not extensively explore long-term generalization or large-scale deployment scenarios. Future research should examine scalability to larger foundation models, robustness under distribution shifts, and alternative human-centered evaluation frameworks. Overall, the study concludes that preference optimization can be reframed as a supervised learning problem, making DPO a

simpler yet theoretically grounded and practically effective alternative to conventional RLHF pipelines.

2.5.28. Large Language Models are Human-Level Prompt Engineers

According to Zhou et al. (2023), high-performing prompts for large language models (LLMs) are typically handcrafted despite strong evidence that task accuracy is highly prompt-dependent. To reduce this gap and reliance on human trial-and-error, they proposed Automatic Prompt Engineer (APE), which frames instruction writing as a program-synthesis-like black-box optimization problem; an LLM proposes a pool of candidate natural-language instructions, and a separate LLM is used to score candidates via zero-shot task performance under a chosen objective. Across extensive experiments, the generated instructions using APE performed better than the previous LLM baseline and were on par with or better than human-written instructions on all 24 Instruction Induction tasks and most BIG-Bench tasks (17/21), while also improving few-shot learning by taking into account optimized instructions to standard in-context prompts and finding better zero-shot chain-of-thought triggers as well as prompts that steer outputs toward truthfulness and/or informativeness. However, they noted significant limitations in the sensitivity of outcomes to meta-prompt design and in the transferability of generated instructions across different execution models, which they suggest that researchers work on for future studies.

2.5.29. Heuristics in multi-criteria decision-making: The cost of fast and frugal decisions

According to Methling et al. (2022), there is a gap concerning how often heuristic simplifications lead to outcomes that diverge from those produced by comprehensive

decision frameworks. To address this, the authors apply various heuristics to a dataset of 945 real personal decisions, using both simulations and experimental comparisons to assess differences between heuristic versus exhaustive decision strategies. Their findings reveal that heuristic methods result in alternative recommendations differing from structured decision processes in over 60 % of cases, with an associated mean relative utility loss of 34.58 % when recommended alternatives diverge, thus highlighting potential costs of information reduction in decision quality. Recognizing the limitations of heuristics, the authors recommend that future research focus on refining heuristic models to mitigate utility loss and investigate contexts in which specific heuristic strategies may be more appropriate or adaptive relative to analytic methods

2.5.30. Fantastically Ordered Prompts and Where to Find Them: Overcoming Few-Shot Prompt Order Sensitivity

Lu et al. (2022) investigated how the ordering of in-context examples influences the performance of few-shot prompting in large language models. The study systematically evaluated all possible permutations of example orderings in few-shot prompts across multiple GPT-2 and GPT-3 model variants and introduced an entropy-based heuristic approach that ranks candidate prompt orders using model-generated predictions on an unlabeled probing set. The findings revealed that prompt ordering alone can lead to substantial performance variability, ranging from near-supervised accuracy to near-random outcomes, even as model size increases. Their proposed entropy-based selection strategies, particularly the GlobalE metric, achieved an average relative performance improvement of approximately 13 percent

across eleven text classification datasets while also reducing variance compared to random or fixed ordering. The study highlights that prompt construction is inherently unstable when treated naively and that structural decisions substantially affect outcomes. The authors recommend framing prompt design as a structured search-and-selection problem and applying heuristic scoring functions to identify robust prompt configurations without relying on labeled validation data.

2.5.31. LoRA: Low-Rank Adaptation of Large Language Models

Hu et al. (2021) introduced LoRA, a parameter-efficient fine-tuning method that injects trainable low-rank matrices into transformer layers while freezing pretrained weights. Their experiments demonstrated reductions of up to 10,000× in trainable parameters compared to full fine-tuning, while maintaining comparable performance across multiple language modeling tasks. The method significantly reduced memory and storage requirements. However, the study concentrated on parameter adaptation efficiency and did not evaluate how prompt structure influences downstream performance. The authors recommend further exploration of optimization strategies that complement parameter-efficient training.

2.5.32. PromptWizard: The future of prompt optimization through feedback-driven self-evolving prompts

Nambi and Ganu (2024) introduced PromptWizard, a feedback-driven prompt optimization framework that improves prompt performance through iterative generation, evaluation, and refinement. The system automatically produces candidate prompts, evaluates their effectiveness based on task performance, and updates them

accordingly, allowing both instructions and in-context examples to be optimized without extensive manual effort. The study demonstrates that this approach can achieve strong performance across multiple tasks while also improving efficiency in terms of token usage and API cost compared to baseline methods. However, the framework primarily focuses on performance-based feedback and iterative improvement, without incorporating a structured multi-criteria evaluation of prompt quality or integrating model-level fine-tuning techniques, indicating an opportunity for future work to develop a more unified and systematic optimization pipeline.

2.6. Synthesis of Local and Foreign Studies

Local research consistently highlights the rapid adoption of artificial intelligence in the Philippines alongside a clear lack of structured frameworks (Espartinez, 2025). Quantitative and qualitative studies across educational and commercial settings show that users heavily rely on AI tools, but their trust and engagement depend strictly on the perceived accuracy of the system (Moreno et al., 2025; Amoozadeh et al., 2024). Filipino students often face redundant and inaccurate AI responses in their academic tasks (Cabuquin et al., 2024). Furthermore, AI use is already normalized across different student year levels (Dela Rosa et al., 2024) , and educators currently treat prompt design as an intuitive practice rather than a structured and evaluated process (Hatmanto et al., 2024). In the industry sector, studies reveal a digital divide where AI adoption is low and constrained by limited infrastructure (Quimba et al., 2024; Hernandez et al., 2023). Both students and institutions exhibit only moderate AI readiness and face an urgent need for guided AI policies (Asirit & Hua, 2023). To address these inaccuracies, foreign studies

shift their focus to technical solutions, such as optimizing prompts to align models and improve instruction following without altering their core training (Cheng et al., 2024; Zhou et al., 2023). These findings suggest that beyond system-level optimization, there is a critical need to guide users toward structured prompting practices, as improper input formulation remains a key source of hallucinations and unreliable outputs (Yao et al., 2024; Cabuquin et al., 2024).

To automate prompt improvements, foreign researchers have developed various optimization techniques, but these methods carry significant limitations. Many studies propose genetic algorithms and evolutionary computation to iteratively mutate and select the best prompts (Hidalgo et al., 2025; Loss & Dhuvad, 2025; Guo et al., 2024). While these approaches improve semantic fidelity, authors explicitly warn about their severe computational costs, high evaluation overhead, and complexity in handling large language model queries (Sécheresse et al., 2025; Poyatos et al., 2024; Hidalgo et al., 2025). Other automated techniques, such as using models as optimizers or relying on textual gradients, also suffer from high API costs and sensitivity to initialization (Pryzant et al., 2023; Yang et al., 2024; Ye et al., 2023). Furthermore, approaches using traditional reinforcement learning are complex and require fine-grained adaptation to prevent instability (Chen et al., 2023). In evaluating these models, relying on fast heuristic decisions instead of comprehensive frameworks can lead to a massive utility loss of over 34 percent, proving that simple evaluation shortcuts are costly (Methling et al., 2022). Additionally, structural choices like prompt ordering or landscape ruggedness can drastically alter outcomes, requiring robust selection strategies (Hintze, 2025; Lu et al., 2022).

These documented limitations validate the need for a highly efficient and mathematically grounded methodology. Direct Preference Optimization solves the instability and high costs of traditional reinforcement learning by reformulating alignment into a simple, stable supervised learning objective (Rafailov et al., 2023; Xiao et al., 2025). Studies confirm that this method effectively refines structured outputs without the computational overhead of previous models (Miao et al., 2024). To further reduce hardware requirements, Quantized Low-Rank Adaptation enables parameter-efficient fine-tuning that performs within 1 to 2 percent of full-precision models while cutting memory consumption significantly (Choi et al., 2025; Hu et al., 2021). It also prevents catastrophic forgetting better than full fine-tuning (Biderman et al., 2024). Finally, since prompt optimization requires balancing multiple competing objectives, a Multi-Criteria Decision-Making framework provides a structured evaluation to prevent the utility loss seen in naive heuristic methods (Methling et al., 2022; Zhao et al., 2025). Combining these methods directly addresses the local gap in unstructured AI use by offering a resource-conscious, stable, and multi-objective system for automatic prompt optimization.

Chapter 3

METHODOLOGY

This chapter outlines the methods, processes, and techniques used to accomplish the objectives of the study. By outlining the following detailed methodologies, this chapter ensures the transparency and reproducibility of this study.

3.1. Type of Research

This study incorporates a Hybrid Developmental and Prescriptive Research approach. According to Roxas & Cawi (2023), developmental research is a process with the aim of establishing a factual basis for the creation of new tools that focus on improvement. The research is grounded in a developmental process that documents the stages of design, development, improvement, and testing. This systematic approach provides empirical evidence of the system's efficacy, demonstrating an advancement in prompt optimization by addressing real-world challenges such as prompt brittleness. Furthermore, the system adopts a prescriptive framework and suggests actionable strategies (Awe, 2024), which are the optimized prompts from the user input. By identifying the raw and optimized states of the user prompts, this study will objectively assess the efficacy of the DPO-aligned model regarding its role in overcoming the problem of prompt brittleness while also enhancing the efficacy of the output.

3.2. System Design

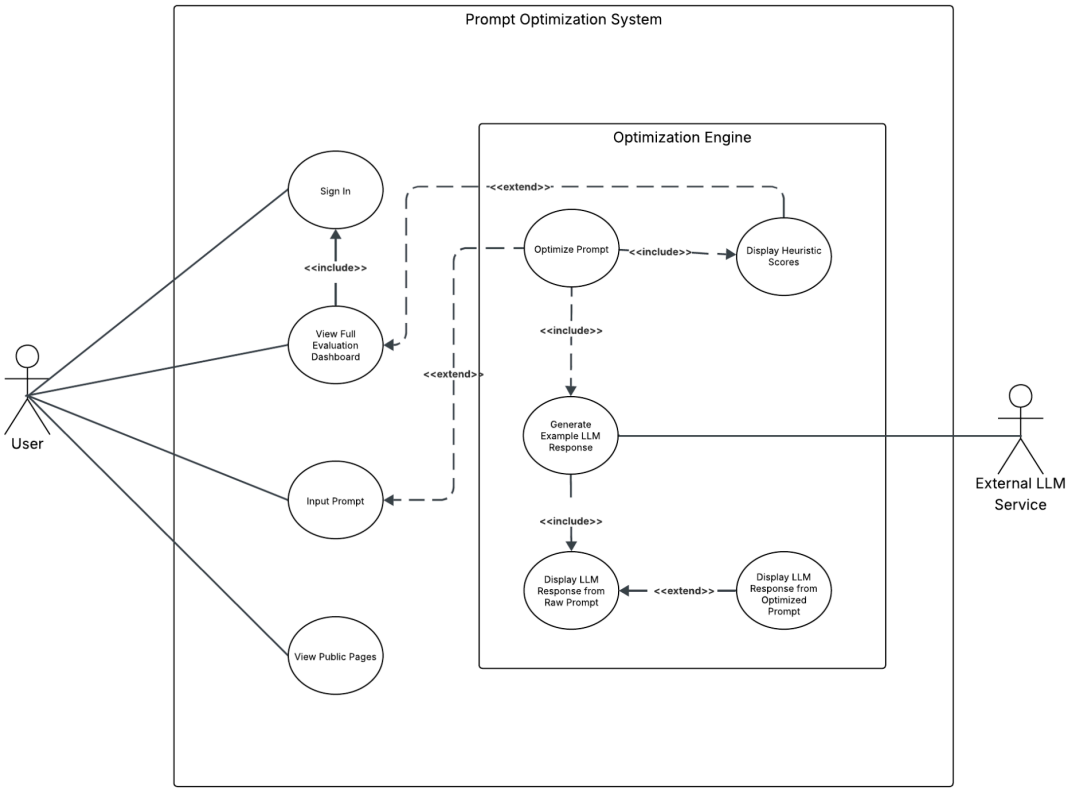

Figure 5. Use Case Diagram of the system

The system follows the use case diagram shown in Figure 5. The design implements a two-actor environment where the primary actor is the User—who initiates requests, and the secondary actor is an External LLM Service responsible for downstream task comparison. The workflow starts when the user interacts with the Input Prompt function. From this step, the user can optionally trigger the core processing within the dedicated Optimization Engine via the Optimize Prompt function, where the internal transformer performs the rewrite. This optimization process mandatorily establishes a quality scoring through the Display Heuristic Scores. For visualizing the results, the optimization also

calls the Generate Example LLM Response to communicate with the secondary actor. This automatically triggers the Display LLM Response from Raw Prompt, which is then extended by the Display LLM Response from Optimized Prompt to provide a side-by-side comparison. Furthermore, access to the View Full Evaluation Dashboard is restricted by an included Sign In feature, ensuring that only logged-in users are able to view the detailed metric snapshots. Additionally, the dashboard's functionality is extended by the Display Heuristic Scores feature, allowing users to optionally view the specific mathematical evaluations directly from their dashboard interface.

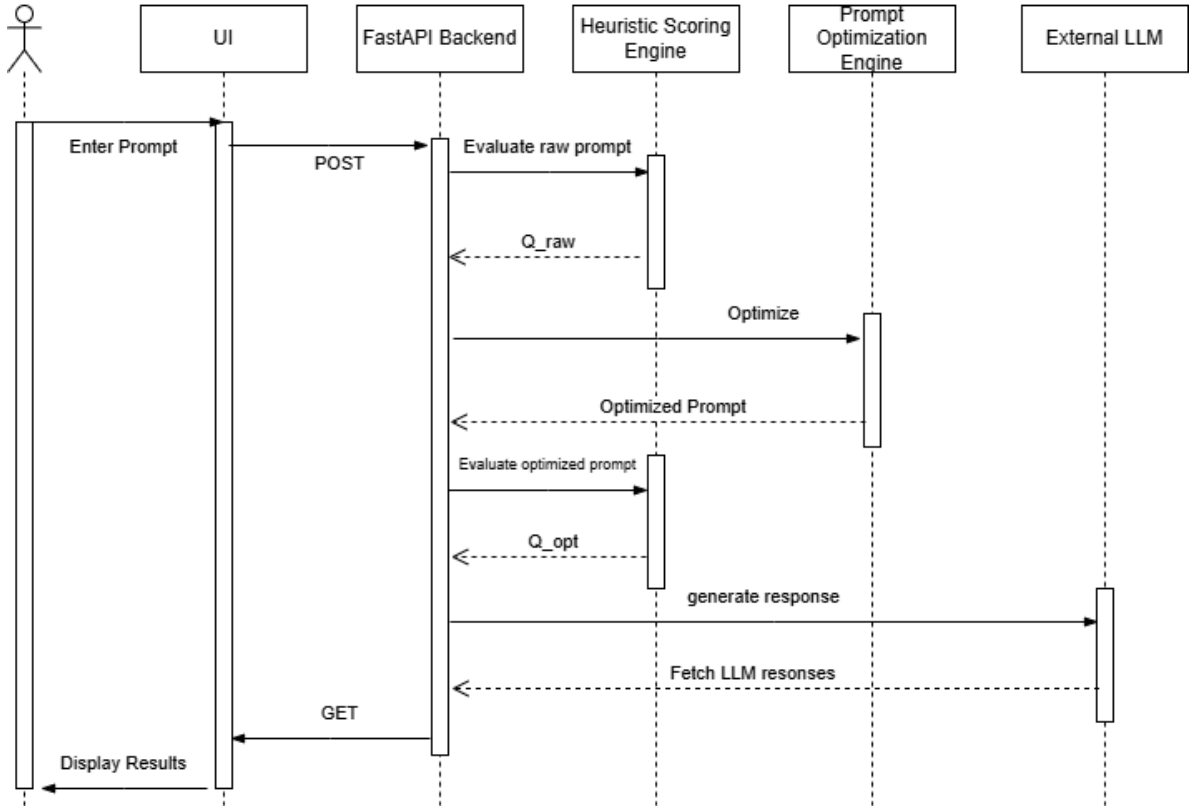

Figure 6. Sequence Diagram

The system follows the sequence diagram shown in Figure 6. The system initiates the system operations when a user inputs a raw prompt and requests for prompt optimization. When the user types and submits a raw prompt, the frontend sends a POST request to the backend. Upon receiving the request, the backend first calls the Heuristic Scoring Engine to evaluate the raw prompt using the defined multi-criteria metrics. The engine returns a numeric quality score, Q-raw. After this initial evaluation, the backend forwards the raw prompt to the Prompt Optimization Engine, this time, which generates a rewritten and optimized version of the prompt. The optimized prompt is then returned to the backend. Next, the backend forwards the optimized prompt to the Heuristic Scoring Engine for post-validation. The engine evaluates the rewritten prompt using the same criteria and returns a heuristic score, Q-opt. Furthermore, the backend now calls an external LLM to generate example output responses using both the raw and rewritten prompts. These responses are retrieved and, together with the heuristic scores and optimization results, stored in the backend. Finally, a result object is sent to the frontend, and the UI displays the raw prompt, the optimized version, the heuristic scores, and example generated LLM responses for downstream comparison and user inspection.

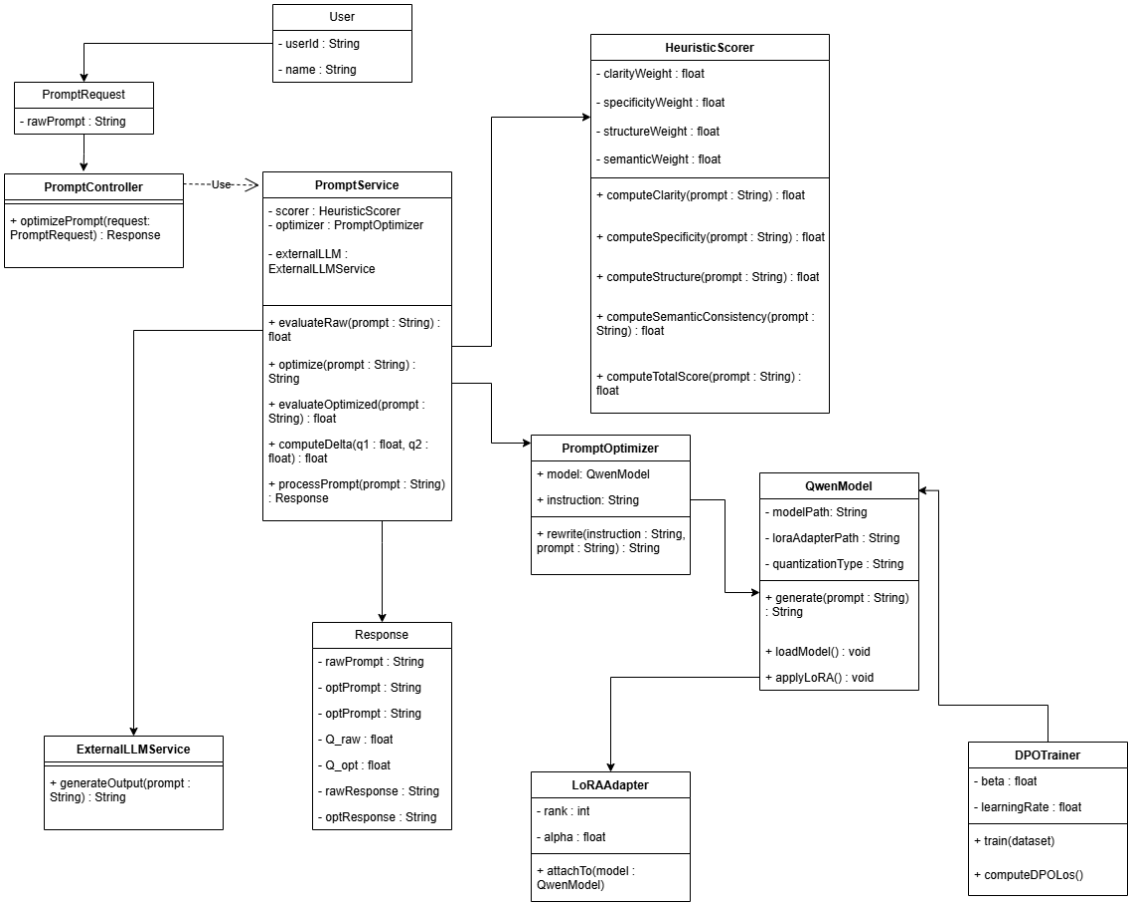

Figure 7. Class Diagram

The system follows the class diagram shown in Figure 7. The diagram maps out the object oriented architecture necessary to support the runtime prompt rewriting and offline model training. Initially, the PromptController accepts a prompt request, passing the workload to the central service orchestrator which is the PromptService. This orchestrator first calls the HeuristicScorer, evaluating the input against the given metrics. After that, the prompt is passed to the PromptOptimizer class. Using the QwenModel to rewrite the prompt, guided by an instruction. The architecture also incorporates model fine tuning,

the QwenModel will be designed to interface with a LoRAAdapter, which will handle the parameter efficient tuning without the need for training all parameters of the entire model. This integration is supervised by the DPOTrainer, which manages the offline training by processing datasets and computing the DPO loss. Finally, the service collects raw and optimized prompts, their comparative scores, and the output from the ExternalLLMService into a comprehensive Response object for user display and validation

3.3. System Architecture

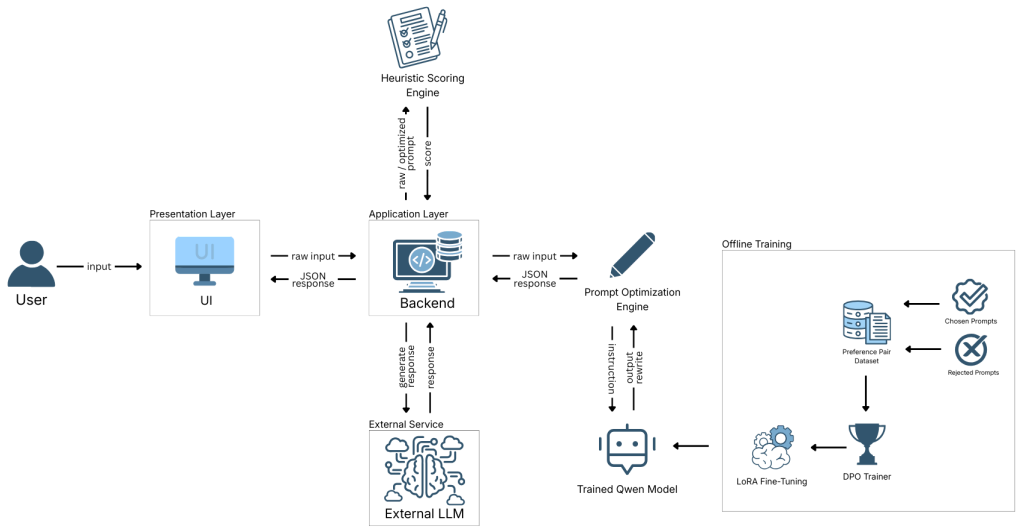

Figure 8. Modular Prompt Optimization System Architecture

The study involves using an established model to be fine-tuned for rewriting prompt structures. Figure 8 demonstrates a layered modular system architecture for this specific task. During runtime, the Presentation Layer (UI), accepts a string input from the user

and forwards it to the Application Layer (Backend). The optimization process starts with the backend calling the Heuristic Scoring Module to compute a score for the inputted prompt, then forwards that prompt to the Prompt Optimization Engine to generate a structured rewrite. It uses the model layer, the fine-tuned Qwen-Instruct Model with LoRA adapters for its processes. Next, the backend performs post-scoring to the optimized prompt in order to produce another score necessary for computing the total improvement score. Then, the system calls an external LLM to produce sample responses for both the raw and optimized prompts. Finally, all the stored values are then returned to the UI for display and user comparison. The boxed Offline Training area is a separate entity from the runtime operation. It shows how pairs of chosen and rejected prompts are selected into the preference dataset and used for fine tuning the model using DPO and QLoRA. This is responsible for making the model updates explicit and offline rather than a continuously running cloud process. This architecture aims to support a reproducible offline learning, that keeps the runtime lightweight and deterministic for user interactions.

3.4. Hardware and Software Requirements

Determining the minimum and recommended device specifications is essential to ensure that the proposed system and the model used would execute efficiently and deliver optimal performance. Defining these requirements is crucial for preventing potential hardware and software limitations that may affect the whole system operation.

Table 1. Hardware and Software Requirements for Development

Category	Minimum	Recommended
GPU	RTX 2080 Ti	RTX 3060 (12GB)
RAM	8GB DDR4	16GB DDR4
Storage	250 GB NVMe SSD	1 TB NVMe SSD
Operating System	Windows 10	Ubuntu 22.04 LTS
Python	Python 3.11	Python 3.11
CUDA / Drivers	CUDA 11.7 / 11.8 (matching PyTorch & bitsandbytes build)	CUDA 11.8 or 12.x, depending on PyTorch/bitsandbytes compatibility; NVIDIA driver matching CUDA
PyTorch	PyTorch 2.x (CUDA-enabled)	PyTorch 2.x (CUDA-enabled)
Core Libraries	transformers, peft, trl (DPO), bitsandbytes, datasets, sentence-transformers, spaCy	transformers, peft (LoRA), trl (DPO), bitsandbytes, accelerate, datasets, sentence-transformers, spaCy

Table 2. Hardware and Software Requirements for Deployment

Category	Minimum	Recommended
CPU	Intel Core i5 - 7th Gen	Intel i9
RAM	8GB DDR4	16GB DDR4
Storage	250 GB NVMe SSD	1 TB NVMe SSD
Operating System	Windows 10	Ubuntu 22.04 LTS
Application Logic(Backend)	FastAPI, Uvicorn/Gunicorn	FastAPI + Uvicorn/Gunicorn
User-Interface(frontend)	React	React, Nginx reverse proxy

Web Browser	Modern web browser	Modern web browser

The technical requirements consist of two parts: development and deployment. The development phase uses a GPU-based stack, including CUDA and various machine learning libraries, to handle the complex mathematical calculations involved in the refinement of the transformer model. The deployment phase focuses on the web accessibility of the services provided, the system processor is used to manage the FastAPI application logic, and the React-based user interface. Distinguishing the training process from the deployment environment ensures the development of an efficient and responsive pipeline for real-time prompt optimization.

3.5. Methods in Developing Software

3.5.1. Scrum Methodology

Peek (2023) described Scrum as a type of Agile methodology that aims to break projects into smaller, manageable increments called sprints. Each sprint typically lasts from one week to one month and is designed to deliver a functional portion of the project within a short time frame. According to the author, Scrum is particularly suitable for operations that require the timely completion of projects, as it promotes adaptability, efficiency, and continuous progress.

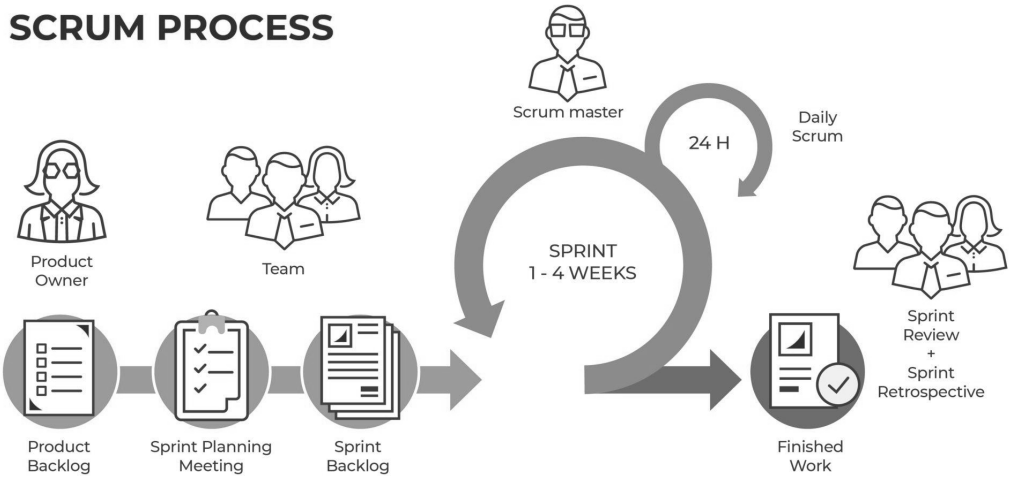

Figure 9. The Scrum Framework

Figure 9 shows the iterative process of the agile scrum framework. Scrum consists of five main components: Sprint, Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective (Owen, 2024). The Sprint serves as the main work period where specific tasks are accomplished. Prior to this, Sprint Planning is conducted to define the sprint's objectives, determine what needs to be done, and outline how the work will be carried out. During the sprint, the team conducts a Daily Scrum, a 15-minute meeting where developers discuss progress, identify obstacles, and ensure they remain on track to achieve the sprint objectives. At the end of each sprint, a Sprint Review is held to present the completed work, gather feedback, and discuss potential improvements or changes. This is followed by the Sprint Retrospective, where the team reflects on the sprint process and identifies strategies for enhancing efficiency and effectiveness. The primary goal of this phase is continuous improvement.

For this study, the researchers adopted the Scrum framework due to its flexible and iterative nature in developing the proposed system. A sprint duration of one week was implemented to maximize efficiency, ensure manageability, and facilitate continuous delivery of outputs within the time constraints of the academic year. The summary of the conducted weekly sprints are as follows:

- Week 1 and 2: **Drafting of Chapters 1 to 3.**
- Week 3 and 4: **Improvement of the UI/UX Wireframe.**
- Week 5: **Finalizing Chapters 1 to 3.**
- Week 6: **Developing the Heuristic Scoring Engine.** This involves programming the multi-criteria metrics for clarity, specificity, and semantic preservation
- Week 7: **Curating the Preference Pair Dataset.** Utilizing the newly built Heuristic Scoring Engine to autonomously score and rank raw prompts versus candidate rewrites to generate data for the DPO loss function.
- Week 8: **Executing the Offline Training Phase.** Fine-tuning the Qwen-Instruct base model using QLoRA for memory efficiency and aligning it using Direct Preference Optimization (DPO).
- Week 9: **Backend Integration.** Connecting the fine-tuned model, the Heuristic Scoring Engine, and the external LLM service through the FastAPI backend
- Week 10: **Frontend Development.** Building the React-based user interface, specifically the prompt input mechanics and the comparative evaluation dashboard

- Week 11: **System Testing.** Performing White Box testing (unit and integration tests for algorithmic components) and Black Box testing for overall system functionality.
- Week 12: **Deployment & Evaluation.** Conducting the final system evaluations using the ISO 25010 framework, Technology Acceptance Model (TAM) surveys, LLM-as-a-judge, and Human Expert reviews

3.5.2. Algorithm

This section describes the primary algorithms and mathematical formulations that will serve as the foundation of the automatic prompt optimization pipeline. The system will rely on a sequence of three major algorithms: QLoRA for memory efficient fine-tuning, MCDM Heuristics for deterministic prompt evaluation, and DPO for aligning the model's output for rewrites.

3.5.2.1. Quantized Low-Rank Adaptation (QLoRA)

Fine-tuning LLMs traditionally requires updating all of its parameter space, necessitating a large computational overhead. LoRA presents a way of mitigating this concern by freezing the pre-trained model weights and injecting rank decomposition matrices, called adapters into each transformer layer (Hu et al., 2021). QLoRa is an extension of this technique. By quantizing these frozen base weights to a 4-bit NormalFloat (NF4) instead of the 16-bit, QLoRA significantly reduces the training's memory footprint without sacrificing too much performance (Dettmers et al., 2023).

Mathematically, the forward pass in a standard LoRA/QLoRA module is defined by this equation:

$$W' = W_0 + B\,A$$

Where:

W' is the updated weight matrix

$W_0 \in \mathbb{R}^{d\,x\,k}$ represents the frozen, 4-bit quantized pre-trained weights

$B \in \mathbb{R}^{d\,x\,r}$ *and* $A \in \mathbb{R}^{r\,x\,k}$ are the trainable low-rank adapters

r is the rank, where $r \ll min(d, k)$

During backpropagation, gradients are only computed for the smaller matrices A and B, which significantly reduces the size of the optimizer states.

```
# QLoRA Forward Pass Pseudocode

def qlora_forward(x, W_4bit, lora_A, lora_B, scale, zero_point):

    # 1. Dequantize base weights on-the-fly
    W_dequant = dequantize(W_4bit, scale, zero_point)

    # 2. Base model computation
    base_output = x @ W_dequant

    # 3. LoRA computation (full precision)
    lora_output = (x @ lora_A) @ lora_B

    # 4. Combine outputs
    return base_output + lora_output


# Memory savings: Base model uses 4x less memory!
```

Figure 10. QLoRA Forward Pass Pseudocode

Note. Reprinted from Dubey, A. (2021). *Advanced PEFT: QLoRA, DoRA & Modern Techniques*.Github.io.https://profitmonk.github.io/visual-ai-tutorials/advanced-peft.html

Figure 3.5.2.1 illustrates the forward computation in QLoRA. Initially, the base model weights are stored in a 4-bit format to save memory, and are temporarily dequantized so they can be multiplied with the input. In parallel, the LoRA adapters (low-rank matrices A and B) compute an additional adjustment in full precision. Finally, the output from the frozen base model and the output from the trainable LoRA adapters are added together. This approach allows the model to learn new knowledge efficiently while using significantly less memory (Dubey, 2021). In this pipeline, QLoRA will be applied to fine-tune the base Qwen-Instruct Model. Since typical consumer hardware consists of physically constrained resources, the 4 bit quantization will allow the large parameter space of the Qwen-Instruct model to be loaded into that constrained memory. The adapters A and B, will serve as the trainable parameters where the knowledge of optimizing prompts will be embedded.

3.5.2.2. Deterministic Heuristics in Multi-Criteria Decision-Making (MCDM)

Figure 11. Steps In MCDM

Note. Reprinted from Taherdoost, H., & Madanchian, M. (2023). Multi-Criteria Decision Making (MCDM) Methods and Concepts. *Encyclopedia*, *3*(1), 77–87. https://doi.org/10.3390/encyclopedia3010006

To evaluate the generated text without relying on another computationally heavy Neural Reward Model, the system will utilize heuristics grounded in MCDM (Methling et al., 2022). Heuristics also is a method of reducing computational complexity by evaluating alternatives, in this context, the rewritten prompts, based on a selected number of defined attributes. Methods utilizing this technique ensure that the final selection is objective, consistent, and justifiable based on the established quantitative standards (Taherdoost & Madanchian, 2023). To implement this paradigm for evaluating the prompts, the system will adopt the three-step process workflow of MCDM illustrated in Figure 11. The process will begin with the first phase: Identify and Select Criteria. For this study, these criteria are defined as Clarity, Specificity, (Ekin, 2023; White, 2023) and Semantic Preservation (Zhu, 2026) selected for their theoretical consistency in determining prompt quality. Then unto the next phase, Determine the Weights of Resources, assigning the relative importance values to each defined criterion based on their impact on the prompt's quality. The final phase includes using the weighted attributes with the deterministic heuristic algorithm to rank the resources—which are the prompt rewrite candidates, calculating composite scores to distinguish chosen and rejected prompts for the construction of preference pairs, and later for evaluating the prompt quality scores.

The evaluation algorithm calculates a composite heuristic score, $H(x)$, for a given prompt x, mathematically expressed as a weighted sum of its criteria:

$$H(x) = w_c C(x) + w_s S(x) + w_p P(x, y)$$

Where:

$C(x)$ is the Clarity score, derived from syntactic dependency parsing.

$S(x)$ is the Specificity score, derived from identifying constraints.

$P(x, y)$ is the Semantic Preservation score, acting as a boundary constraint to ensure the original intent of the raw prompt (x) is retained.

w_c, w_s, w_t, w_p are the designated weights for each criterion.

To ensure that the optimized prompt has not hallucinated or deviated from the raw prompt's original intent, Semantic Preservation $P(x, y)$ will utilize BERTScore Recall (R_{BERT}). By generating contextual embeddings for every individual token using a pre-trained language model. BERTScore then employs greedy matching to find the maximum cosine similarity between each token in the original raw prompt (x) and the tokens in the optimized prompt (y).

This deterministic heuristic algorithm will serve as the autonomous scoring engine within the data generation phase, and later for the prompt quality scoring. When the system generates prompt rewrites, this algorithm will parse for Clarity, Specificity, and Structure markers (White et al., 2023) while computing BERTScore Recall to enforce strict semantic boundaries. By isolating Recall, the system ensures that the generated rewritten

prompt has not removed important details from the raw prompt's original intent. By ranking composite scores $H(x)$, the system will fill out the necessary data into the chosen prompts (y_w) for creating the preference pair dataset without further human intervention, which will then be passed to the DPO trainer.

3.5.2.3. Direct Preference Optimization (DPO)

```
import torch.nn.functional as F

def dpo_loss(pi_logps, ref_logps, yw_idxs, yl_idxs, beta):
    """
    pi_logps: policy logprobs, shape (B,)
    ref_logps: reference model logprobs, shape (B,)
    yw_idxs: preferred completion indices in [0, B-1], shape (T,)
    yl_idxs: dispreferred completion indices in [0, B-1], shape (T,)
    beta: temperature controlling strength of KL penalty

    Each pair of (yw_idxs[i], yl_idxs[i]) represents the
      indices of a single preference pair.
    """

    pi_yw_logps,  pi_yl_logps =  pi_logps[yw_idxs],  pi_logps[yl_idxs]
    ref_yw_logps, ref_yl_logps = ref_logps[yw_idxs], ref_logps[yl_idxs]

    pi_logratios  = pi_yw_logps - pi_yl_logps
    ref_logratios = ref_yw_logps - ref_yl_logps

    losses = -F.logsigmoid(beta * (pi_logratios - ref_logratios))
    rewards = beta * (pi_logps - ref_logps).detach()

    return losses, rewards
```

Figure 12. DPO Implementation Details and Hyperparameters

Note. Reprinted from Rafailov, R., Sharma, A., Mitchell, E., Ermon, S., Manning, C., Finn, C., & Cz Biohub. (2024). *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*. https://arxiv.org/pdf/2305.18290

DPO is a stable and efficient algorithm for fine-tuning LLMS to align with human preferences, bypassing the complexity of traditional Reinforcement Learning from Human Feedback (RLHF). Introduced by Rafailov et al. (2023), DPO leverages a theoretical insight that the reward model in RLHF can be analytically reparameterized in terms of the optimal policy itself. Unlike RLHF, which requires training a separate reward model and then optimizing a policy using reinforcement learning algorithms like Proximal Policy Optimization (PPO), DPO sets the alignment process as a straightforward classification task, optimizing the model directly based on preference pairs. This approach significantly simplifies the training pipeline, reduces computational overhead, and improves convergence stability compared to RL-based methods. The objective is to minimize the DPO loss function, which increases the relative log probability of the preferred response over the rejected response, scaled by a temperature parameter:

$$L_{DPO}(\pi_\theta; \pi_{ref}) = - E_{(x,y_w,y_l)\sim D} [log\, \sigma (\beta\, log \frac{\pi_\theta(y_w | x)}{\pi_{ref}(y_w | x)} - \beta\, log \frac{\pi_\theta(y_l | x)}{\pi_{ref}(y_l | x)})]$$

To further simplify the implementation of this theoretical framework practically within the active training loop, the system translates the mathematical operations into a python function, as shown in Figure 12.

Where:

π_θ is the active, trainable language policy.

π_{ref} is the frozen reference model.

x is the raw input prompt.

y_w and y_l are the chosen and rejected prompts in the dataset

σ represents the logistic function

β is the KL-divergence penalty parameter controlling the deviation from the reference model

DPO handles the mathematical foundation for optimizing the Qwen-Instruct model using the generated preference pair dataset. Once the MCDM Heuristic Engine has curated a dataset of raw prompts mapped to chosen and rejected pairs, this dataset is then fed into the DPO loss function. By utilizing the maximum likelihood, the base Qwen-Instruct model will have its policy π_θ updated using QLoRA adapters so it favors the generation of a more structured prompt that satisfies the criteria of the attributes from MCDM, without the overhead of training a separate reward model.

3.6. Methods in Evaluating the System

3.6.1. ISO 25010 Quality Evaluation

To ensure the system meets the standards of software engineering, the researchers complied with the ISO 25010 framework, which directly aligns with the study's objective of reducing prompt brittleness and enhancing output quality:

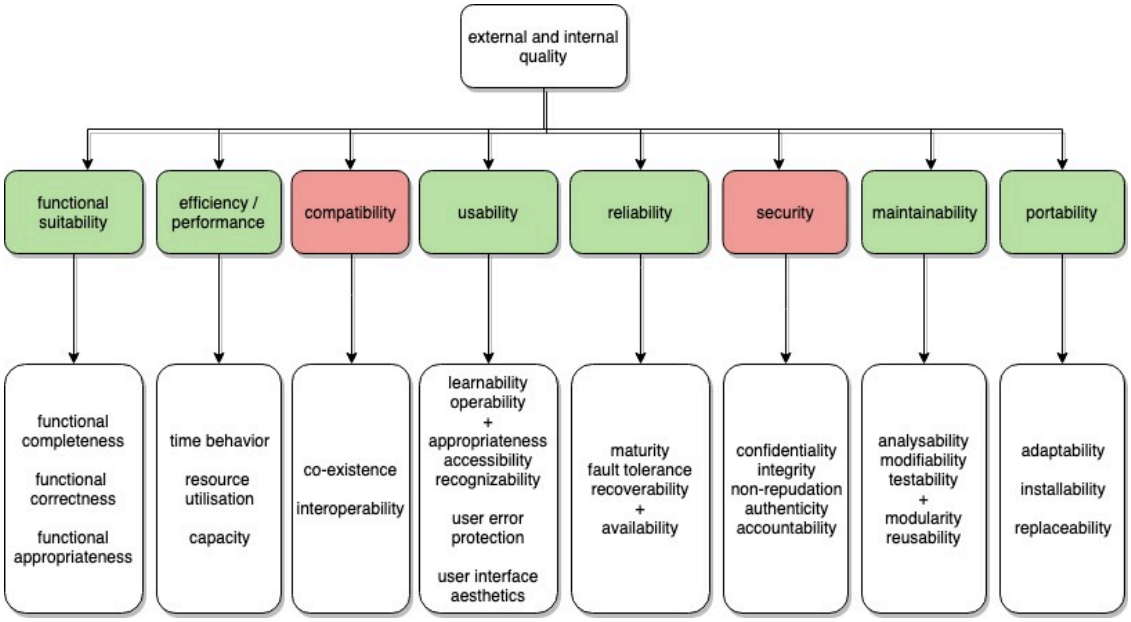

Figure 13. ISO/IEC 25010 Categorization of Software Quality Requirements

As shown in Figure 13, the researchers applied the ISO 25010 System and Software Quality Model in evaluating the technical and quality-related aspects of the system. The ISO 25010 System and Software Quality Model is a structured categorization of software quality requirements. The application of this model ensures that the evaluation is based on established software engineering principles, which enables an objective evaluation of the system's ability to successfully improve output quality.

3.6.2. TAM-Based User Acceptance

Complementing the technical ISO standards, the Technology Acceptance Model (TAM) is used to measure the behavioral intention of users to adopt the technology (Koteczki & Balassa, 2025). Participants will evaluate if utilizing the DPO-aligned

optimization engine significantly improves their workflow efficiency and the quality of their LLM-generated code.

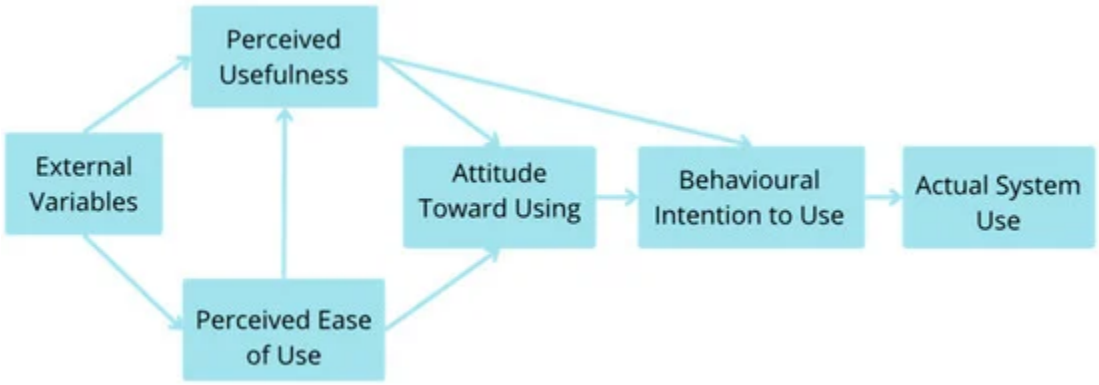

Figure 14. Technology Acceptance Model (TAM)

As shown in Figure 14, this would measure the extent to which a user believes using the system would be effort-free, focusing on the simplicity of the prompt-requirement capture process. It also quantifies the likelihood that the user will integrate the system into their regular development or prompting routine.

3.6.3. BERTScore - Recall

As an intent metric, BERTScore-Recall is employed. As cited by Slobodianiuk & Semerikov (2025), while the traditional metrics like BLEU and ROUGE are still widely used, there has been a significant increase in interest in BERTScore due to the massive increase in the use of neural text generation. One of the uses of BERTScore can be identified in the study conducted by Ansari et al. (2026), where they employed a multi-objective prompt learning framework for English-Urdu summarization tasks.

In their research, the performance is calculated by finding the ROUGE-L score along with BERTScore to determine the semantic meaning. Ansari et al. (2026) were able to perform a more accurate semantic similarity evaluation than the n-gram approach. Additionally, Ortolan (2023) explains that BERTScore uses the BERT (Bidirectional Encoder Representations from Transformers) model to represent the text in contextual embeddings, which focus on the meaning and context of words and phrases as opposed to keyword matching.

$$R_{BERT} = \frac{1}{|x|} \sum_{x_i \epsilon x} max_{\hat{x}_j \epsilon \hat{x}} x_i \top \hat{x}_j$$

Where:

$x = candidate\ sentence\ tokens$

$\hat{x} = reference\ sentence\ tokens$

$|x| = total\ number\ of\ reference\ tokens$

$x_i = contextual\ embedding\ of\ the\ i_{th}\ token\ in\ the\ reference$

$\hat{x}_j = contextual\ embedding\ of\ the\ j_{th}\ token\ in\ the\ candidate$

$max_{\hat{x}_j \epsilon \hat{x}} = greedy\ matching\ operation$

BERTScore can be classified into 3 variations: Precision, Recall, and F1-measure. For this research, Recall is chosen as a measurement method to assess the retention of the raw prompt's semantics or original intent. If Precision or F1-measure were included, it would yield inconsistent results. These two methods focus on the structural accuracy of a text, which would be completely altered by the system in the optimization process.

3.6.4. LLM-as-a-Judge

The study will utilize various LLMs such as ChatGPT, Gemini, and Deepseek to act as judges in the study to maintain the objectivity of the results and to evaluate whether the system would actually optimize the raw input. Similar to the study done by Cheng et al. (2024), the LLM judges will be presented with a blind, pairwise comparison containing both the raw user prompt and the system-optimized prompt. The models will be instructed to evaluate both versions and select a winner based on which is structurally better overall to calculate the win-rate. A high-performing LLM judge has the capability of attaining more than 80% agreement with human preferences and can attain the same level of consistency as human experts in the MT-Bench benchmark (Zheng et al., 2023). This shows that it’s possible to attain the results without having to involve humans in the process in order to avoid exhaustion bias.

3.6.5. Human Expert Evaluation

A Human Expert Evaluation will be used in this study. Blind testing will be employed in this step to ensure an objective evaluation of the outputs and reduce bias that could arise in the LLM-as-a-judge. A total of ten model outputs are to be examined by nine subject matter experts. Five raw user prompts and five system-optimized prompts are used to create these samples, which are randomized and de-identified to prevent the evaluators from learning about the source system. The experts will assess the outputs using a 5-point Likert scale. A weighted mean analysis will then be applied to the combined scores to ensure a fair statistical comparison of the raw and optimized prompts' performance.

3.7. Testing Methods

3.7.1. Black Box Testing

Black box testing is a functional testing method that evaluates whether a system produces correct outputs based solely on given inputs without examining its internal structure. Recent research highlights that black box testing remains essential for validating system functionality and ensuring alignment with user requirements, particularly in academic and educational software systems (Kumar & Mishra, 2022). To evaluate user acceptance, the study will employ a survey instrument grounded in the Technology Acceptance Model (TAM). Whereas TAM remains a reliable framework for predicting user adoption of technologies, particularly through the

constructs of perceived usefulness and perceived ease of use (Al-Emran & Teo, 2021).

3.7.2. White Box Testing

White-box testing, also called structural or glass-box testing, inspects internal code paths, logic, and data flow to verify correctness at the unit and integration levels. For this study, white-box testing should include unit tests and path/branch coverage for the Heuristic Scorer (e.g., metric computations, normalization and weight application), correctness checks for score aggregation, and integration tests that validate data passed between the scorer, optimizer, and persistence layers. White-box techniques (control-flow analysis, boundary tests, and cyclomatic complexity checks) are particularly valuable for algorithmic components because they expose logic errors, off-by-one conditions, and edge cases that black-box tests might miss. Recent reviews of modern software testing emphasize combining structural white-box tests with higher-level functional tests to improve defect detection and maintainability in complex systems (Izzat & Saleem, 2023).

3.8. Data Gathering Procedures

To successfully align the Qwen-Instruct model using Direct Preference Optimization, a custom preference pair dataset must be constructed. This dataset is developed through a combination of user-centered data collection, expert validation, and automated data generation techniques.

3.8.1 Survey Procedures

After the system has been built, the researchers will first secure permission from the academic office of FEU Institute of Technology to conduct the study and distribute survey questionnaires to qualified students. The survey will examine how students use Artificial Intelligence in their academic and daily tasks, as well as their experience in using the proposed prompt optimization system. Its content will be based on the ISO/IEC 25010 System and Software Quality Model and the Technology Acceptance Model (TAM), incorporating items related to usability, functional suitability, reliability, performance efficiency, perceived usefulness, and perceived ease of use. The instrument will undergo pilot testing first to ensure clarity and alignment with the study objectives.

3.8.2 Online Resources

The methodology begins by sourcing initial data from the BPO repository on Hugging Face, extracting only the unstructured raw prompts to serve as the foundational inputs (Cheng et al., 2023). These extracted prompts will then be fed into an external large language model, which will generate a diverse set of candidate rewrites for each input. Subsequently, our proposed multicriteria decision-making heuristic scoring engine will evaluate the generated rewrites. The engine ranks the variations by calculating a composite score derived from clarity, specificity, and semantic preservation. For every original input, the candidate rewrite achieving the highest composite score is designated as the chosen response, and the base raw prompt is designated as the rejected response. This

autonomous scoring process allows us to independently construct the tailored preference dataset necessary to feed the loss function and fine-tune the model.

3.9. Respondents of the Study and Sampling Techniques

The study will employ a stratified purposive sampling approach to select participants equipped with the specific knowledge and experience necessary to evaluate the proposed prompt optimization system. The population will consist of currently enrolled BSCSSE first-year students with profound knowledge in AI tools. In order to establish a statistically representative baseline pool, Slovin's Formula will be applied with a 5% margin of error, to yield a computed minimum sample size of 52 students. Before the actual survey, participants will be asked to complete a preliminary questionnaire to assess their proficiency and confidence levels in using AI tools.

Table 3. Respondents Size Matrix

Type of Respondent	Number of Respondents
First Year Students with Proficiency in using AI Tools	52
Expert Evaluators	9

1. **BSCSSE first-year Students**. A total of fifty-two (52) students will be selected via purposive sampling to participate in the detailed system evaluation. To proceed to the main questionnaire regarding the proposed system, students must meet at least one of two inclusion criteria: possessing an active subscription to a

generative AI platform, or achieving a proficiency and confidence level of seven (7) or more on the preliminary assessment. Establishing this baseline of prior AI experience ensures that the selected respondents are adequately equipped to provide informed, reliable feedback on the system's usability, functionality, output quality, and overall effectiveness.

2. **Expert Evaluators.** Nine (9) expert respondents will be selected to assess the technical accuracy and instructional quality of the generated outputs. Experts must meet at least one of the following criteria: A minimum of two (2) years of professional software development experience; At least one (1) year of experience teaching programming-related subjects; or Active contribution to recognized software development or open-source projects.

This approach ensures statistical grounding while focusing system evaluation on experienced and qualified respondents.

3.10. Statistical Treatment of Data

3.10.1. Weighted Mean

For the analysis of the Likert scale, the researchers will calculate the weighted mean of the responses for each question using the formula:

$$Weighted\ Mean = \frac{\sum_{i=1}^{n} W_i x_i}{\sum_{i=1}^{n} W_i}$$

Where:

$$W_i = Total\ Number\ of\ Respondents\ that\ Selected\ a\ Specific\ Rating$$

$$x_i = Numerical\ Value\ of\ the\ Likert\ Scale\ Rating$$

$$n = Total\ Number\ of\ Likert\ Scale\ Categories$$

The mean will be determined by finding the sum of the rating scores (x_i) multiplied by the response frequency (W_i), divided by the total number of responses for each question. By implementing this formula, the study can find the total consensus of the evaluators with regard to the success of the system. The computed weighted average can be mapped to the Verbal Interpretation scale to find the level of success achieved by the system in conformity with the ISO 25010 & TAM standards.

3.10.2. Likert Scale

The Likert scale will be used as an instrument for quantifying the qualitative perceptions of users concerning the performance of the system. It is a method for measuring qualitative factors like usability or effectiveness in an organized manner by indicating specific numeric values against these perceptions. In this particular study, the 5-point Likert scale will be used to classify the responses, and then the calculated range of averages is used to determine the level of acceptance of the system.

Table 4. 5-point Likert Scale

Scale	Range of Means	Value Interpretation
1	1.00 - 1.79	Strongly Disagree
2	1.80 - 2.59	Disagree
3	2.60 - 3.39	Neutral
4	3.40 - 4.19	Agree
5	4.20 - 5.00	Strongly Agree

This interpretation framework will enable the researchers to translate the numerical averages obtained from the weighted mean into a descriptive perception. Therefore, the success of the performance of the system can be communicated objectively with the help of numerical evidence and qualitative descriptors. This ensures that the evaluation of the system's performance in relation to the ISO 25010 and TAM dimensions is both clear and objective.

3.10.3. Slovin's Formula

To accurately determine the sample size needed for user evaluation, the researchers will applied Slovin's Formula. The target population that will be used in the formula is solely focused on students, specifically those First year students who are enrolled in computer science with proficiency in using AI.

$$n = \frac{N}{1+Ne^2}$$

Where:

$n = Students' \ Mandatory \ Sample \ Size$

N = the student group's total population.

e = Error Margin

Given a total population of 60 students and a desired margin of error of 5%, the required sample size is calculated as follows:

$$n = \frac{60}{1 + 60(0.05)^2}$$

Resulting in a final sample size of **52** participants.

3.10.4. Cronbach's Alpha

According to Sheposh (2024), the purpose of Cronbach's alpha is to give a numerical value for a set of test data's internal consistency. This is used to verify the consistency and reliability of the Likert Scale that will be used in assessing the system's ISO 25010 and TAM compliance. This statistical measure determines how closely related the set of items is as a group, ensuring that the survey yields consistent results.

$$\alpha = \frac{k}{k-1}\left(1 - \frac{\sum_{i=1}^{k} s_y^2}{s_x^2}\right)$$

Where:

$a = Cronbach's\ alpha\ coefficient$

$k = Number\ of\ items\ in\ the\ scale$

$s_y^2 = Variance\ of\ each\ individual\ item$

$s_x^2 = Variance\ of\ the\ total\ scale$

The resulting coefficient will be interpreted using the following standard reliability measurement to ensure the instrument is fit for measuring the qualitative perceptions of the users regarding the prompt optimization system

3.10.5 Win-Rate

Following the methodology of Zhu et al. (2026) and Cheng et al. (2024), Win-Rate analysis provides a numerical representation of the system's consistency in transforming raw user inputs into optimized prompts. This will be used to evaluate the results of a pairwise blind test, where the LLM-as-a-Judge chooses between a raw prompt and its system-optimized version to determine which is better structurally.

$$WR = \left(\frac{w}{T}\right) \times 100\%$$

Where:

$WR = Win\ Rate\ Percentage$

$w = Total\ occurrences\ where\ the\ LLM\ chooses\ the\ optimized\ prompt$

$T = Total\ Evaluation\ Samples$

The resulting percentage serves as an indicator of the system's success, providing a metric to prove whether the optimization process consistently outperforms the original user input in generating high-quality prompts.

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APPENDICES

APPENDIX A

Thesis Title Approval Form

COLLEGE OF COMPUTER STUDIES DEPARTMENT

THESIS PROJECT
TITLE PROPOSAL
(CS0025 – Software Engineering 1)

Group Name: Quad Core	Program: **Bachelor of Science in Computer Science with specialization in Software Engineering**	
Member's Name: 1. Manuzon, Justin Manuel R. 2. Bagayan, Jasper C. 3. Reyes Anneliese Gaille T. 4. San Diego, Christian Gabrielle O.	Term: [ ] 1st [✓] 2nd [ ] 3rd	Academic Year: 2025 - 2026
Proposed Project Title: Prompt Optimization Pipeline Using Multi-Criteria Heuristic Evaluation and Transformer Fine-Tuning Algorithms		

1.0 Area of Investigation:

The field of Requirements Engineering and Human–Computer Interaction (HCI) continues to face challenges related to the precision and structure of natural language inputs. As modern software development increasingly integrates Generative Synthesis Engines—systems capable of producing code, documentation, or textual output from user-provided instructions—the ambiguity inherent in human language becomes a critical bottleneck. The effectiveness of these systems is strongly dependent on the quality and structure of the input prompt, reflecting the principle commonly known as "Garbage In, Garbage Out" (Halilaj et al., 2025).

Although Generative Synthesis Engines have significantly advanced in terms of model capacity and reasoning ability, the responsibility of crafting effective prompts remains largely manual. Users and developers frequently rely on trial-and-error approaches to refine their inputs in order to obtain acceptable outputs (Guo, 2024). This manual refinement process introduces inefficiencies into software workflows and increases cognitive load, particularly for non-expert users.

This project investigates algorithmic prompt optimization as a software engineering problem. By treating prompt formulation as a measurable and improvable process, the study explores how structured evaluation and learning-based techniques can be applied to automatically enhance natural language inputs prior to execution by generative systems. This approach is supported by recent studies indicating that prompt structure and clarity have a direct impact on computational task success (Santana Jr. et al., 2025).

2.0 Background and Rationale of the Project/System:

Generative tools such as automated code assistants, documentation generators, and text summarization systems are now commonly used in software development and general productivity tasks. Despite their widespread adoption, a persistent gap remains between human intent and machine interpretation. Prompts that lack specificity, constraints, or contextual information—such as "fix the error"—often lead to unpredictable or low-quality outputs. In contrast, well-structured prompts produce more accurate and consistent results (Khojah et al., 2024).

At present, there is no standardized software system that automatically evaluates and refines user prompts before they are processed by generative models. Users must repeatedly modify their inputs based on observed outputs, resulting in wasted time and inefficient use of computational resources. This study proposes the development of a Prompt Optimization System that integrates two core components:

1. a multi-criteria heuristic evaluation module that quantitatively measures prompt quality, and
2. a transformer-based prompt optimization model fine-tuned using parameter-efficient techniques such as Low-Rank Adaptation (LoRA) and preference-based learning.

The heuristic evaluation module assesses prompt quality based on interpretable linguistic and semantic criteria, while the transformer model learns to generate optimized prompts that maximize these quality measures. This design shifts prompt refinement from a subjective, manual activity to a systematic, data-driven software process. The system will be implemented as a web-based application using FastAPI for backend services and React for visualization and user interaction.

3.0 Statement of Objectives:

General Objective:

The general objective of this study is to develop a web-based software system that automatically refines natural language prompts by combining multi-criteria heuristic evaluation with transformer-based prompt optimization techniques.

Specific Objectives:

- To develop a Heuristic Scoring Module that evaluates prompt quality using multiple criteria such as clarity, specificity, conciseness, semantic preservation, and constraint presence.
- To implement a Transformer-Based Prompt Optimizer using parameter-efficient fine-tuning techniques that learn to improve raw prompts based on heuristic feedback.
- To design a Prompt Quality Scoring Function that mathematically aggregates multiple heuristic metrics into a single interpretable quality score.
- To develop a Comparative Visualization Interface that presents differences between raw and optimized prompts, including metric-level improvements.
- To evaluate the effectiveness of the system by measuring improvements in prompt quality and reductions in user iteration attempts when interacting with generative tools.

4.0 Scope and Limitations of the Study:

This study focuses on the design and implementation of a prompt optimization pipeline that operates on English-language natural language prompts. The system is intended to optimize prompts for a variety of generative tasks, including documentation generation, logic formulation, and text summarization, without being limited to programming-related prompts.

The core functionality of the system is based on heuristic evaluation and transformer fine-tuning. While the system improves prompt structure and clarity, it does not assess the logical correctness or feasibility of the user's intent. For example, if a user requests an impossible task, the system will optimize the phrasing of the request rather than correct the underlying logic.

The system is implemented as a standalone web application and does not include enterprise-level features such as large-scale authentication, distributed training, or cloud-based deployment. Real-time learning is outside the scope of this study; model fine-tuning is performed offline using curated datasets and heuristic feedback.

5.0 Target Beneficiaries:

To Technical Writers. Can use the optimization engine to ensure technical documentation queries are precise and unambiguous. The system aids in removing linguistic "noise," ensuring that the final output is concise and easily engaging for synthesis engines or human readers.

To General End-Users (Non-Technical). Non-technical users who rely on generative tools for everyday tasks such as drafting emails, summarizing content, or planning activities can benefit from automated prompt refinement without requiring specialized knowledge of prompt engineering.

To Software Engineers and AI Researchers. The system will provide a structured framework for studying the relationship between prompt quality and generative model performance, supporting research into AI-assisted software development and human–AI interaction.

To Computer Science Students. Students can use the system as a practical example of how heuristic evaluation, optimization pipelines, and transformer fine-tuning are applied to real-world software engineering problems.

To Future Researchers. This study serves as a foundational reference for future work on prompt optimization, heuristic evaluation, and learning-based refinement strategies in natural language processing and software engineering contexts.

6.0 SDG Category Number:

SDG 9 – Industry, Innovation, and Infrastructure

This project promotes innovation by advancing the way developers interact with AI-driven systems, improving productivity, and fostering sustainable technology development within the software engineering industry.

7.0 References:

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APPENDIX B

Title

APPENDIX C

Title